

# Hold On to Your Cash:

## CEO Speech Uncertainty and Financing Conservatism

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### Abstract

[Abstract pending Step 6 prose population.]

## I Introduction

Earnings calls are a high-frequency setting where CEOs describe firm conditions to outsiders in their own words. We ask whether CEO speech uncertainty during these calls predicts firm precautionary cash holdings. Our measurement is built on the CEO speech-uncertainty method of [Dzielinski, Wagner, and Zeckhauser \(2021\)](#), which we extend from market reactions, analyst behavior, firm performance, and governance outcomes to firm financing-policy outcomes.

We document that within-firm increases in CEO speech uncertainty during earnings calls predict higher precautionary cash holdings, identifying a real-time signal of firm financing-policy adjustment grounded in the precautionary cash motive ([Opler, Pinkowitz, Stulz, & Williamson, 1999](#); [Bates, Kahle, & Stulz, 2009](#)). The relation is positive in twelve of twelve specifications (six contemporaneous and six one-quarter-ahead) under one-tailed tests at the ten-percent level. The cash response amplifies under both financing-friction stress and cash-flow-volatility stress, consistent with a single precautionary channel triggered by two distinct stress conditions ([Almeida, Campello, & Weisbach, 2004](#); [Han & Qiu, 2007](#)).

External uncertainty proxies—political risk, economic policy uncertainty, and product-market competition—covary with the speech-uncertainty measure in the predicted directions, supporting the measure’s construct validity. Post-call bid-ask spreads widen and SEC conference-call comment letters reference the same speech signal, corroborating the precautionary interpretation through market information-asymmetry and disclosure-insufficiency channels.

Reverse-causality concerns are addressed via cross-sectional *modus tollens* on the two precautionary triggers, and we acknowledge as a limitation that no exogenous-shock or instrumental-variable identification is pursued.

Our main contribution is to establish CEO speech uncertainty during earnings calls as a real-time signal of firm precautionary cash holdings, extending the method beyond its original scope of market reactions, analyst behavior, firm performance, and governance outcomes. The remainder of the paper proceeds as follows. Section II develops the conceptual framework, hypotheses, and empirical strategy. Section III presents the main empirical analyses—cash holdings, the two precautionary triggers, and the endogeneity treatment. Section IV extends to outsider-reaction tests. Section V concludes with limitations and directions for future work.

## II Conceptual Framework and Empirical Strategy

### II.1 Conceptual Framework

Firms hold cash to insure against future funding shortfalls. The precautionary cash motive predicts that firms with riskier cash flows or poor access to external capital hold more cash ([Opler et al., 1999](#); [Bates et al., 2009](#)). The motive operates through a single mechanism: cash buffers allow firms to fund profitable investment without recourse to costly external capital, mitigating underinvestment when internal cash flow falls short of investment needs. Firms most exposed to this risk—those facing the largest gap between expected internal cash flow and expected required investment—optimally hold the largest buffers.

Two stress conditions amplify the precautionary response. The first is financing-friction stress, in which limited debt-market access leads firms to accumulate cash to safeguard future investment (Almeida et al., 2004); the second is cash-flow-volatility stress, in which volatile operating cash flows lead firms to hold larger buffers to smooth funding gaps (Han & Qiu, 2007). The two stresses are conceptually distinct: one constrains the supply of external capital, the other increases the variance of internal capital. Both trigger the same precautionary response, generating two testable amplifications of the cash response under stress.

## II.2 Hypothesis Development

We formalize three hypotheses linking CEO speech uncertainty during earnings calls to firm precautionary cash holdings. The main hypothesis predicts a positive association between speech uncertainty and cash; two amplification hypotheses predict that the association strengthens under each of the two stress triggers identified in Section II.

Speech uncertainty is a candidate real-time signal of the precautionary motive because earnings-call language varies with managerial perceptions of firm conditions (Dzielinski et al., 2021), and CEO-specific traits transmit to firm policy through the speaker's own descriptions (Bertrand & Schoar, 2003). When CEOs perceive heightened firm-level uncertainty, the precautionary mechanism predicts that firms hold more cash to insure against future funding shortfalls (Opler et al., 1999; Bates et al., 2009).

**Hypothesis 1.** *An increase in CEO speech uncertainty during an earnings call is associated with a higher cash-to-assets ratio at the firm.*

The two stress conditions of Section II predict heterogeneous amplification of H1. Financing-friction stress operates through the supply of external capital: rated firms can issue debt at low cost, while unrated firms face binding external-finance constraints and accumulate cash to safeguard future investment (Faulkender & Petersen, 2006; Almeida et al., 2004). Cash-flow-volatility stress operates through the variance of internal capital: firms with volatile operating cash flows hold larger buffers to smooth funding gaps (Han & Qiu, 2007).

**Hypothesis 1a.** *The association between CEO speech uncertainty and cash holdings is stronger for firms facing financing-friction stress (unrated firms) than for firms with debt-market access (rated firms).*

**Hypothesis 1b.** *The association between CEO speech uncertainty and cash holdings is stronger for firms facing cash-flow-volatility stress (high-volatility firms) than for firms with stable cash flows (low-volatility firms).*

## II.3 Estimation of Main Variables

We measure CEO speech uncertainty during earnings calls following Dzielinski et al. (2021). We construct three independent variables from the percentage of uncertainty-words in CEO speech segments: ClarityCEO and UncResCEO are extracted from the unscripted Q&A segment via the DWZ decomposition; UncPreCEO is the raw percentage in the scripted presentation segment.

ClarityCEO is the negative of the CEO entity fixed effect from a regression of Q&A-segment uncertainty on call-level controls (DWZ Eq. 5). It captures the CEO's habitual baseline level of uncertainty-language. UncResCEO is the residual from the same family of regressions (DWZ Eq. 4); it captures within-CEO variation in Q&A uncertainty orthogonal to the baseline. UncPreCEO is the raw percentage of uncertainty-words in the CEO's scripted opening, typically prepared in advance and vetted by investor-relations staff before the call.

Our use of UncPreCEO as an independent variable extends the original method. Dzielinski et al. (2021) use the presentation-segment measure as a control variable in their Q&A residual extraction. We treat UncPreCEO as an independent variable because it may carry an informative signal about firm conditions. We disclose this departure and report results separately for each of the three independent variables in Section III.

## II.4 Methodology

We estimate panel regressions of cash holdings on the three CEO speech-uncertainty variables and a vector of firm-level controls. The dependent variable is the cash-to-assets ratio, measured contemporaneously ( $Cash_t$ ) and at a one-quarter horizon ( $Cash_{t+1}$ ). Each regression includes ClarityCEO, UncResCEO, and UncPreCEO jointly as independent variables of interest.

For each dependent-variable horizon, we report a six-cell fixed-effects ladder: (i) industry fixed effects with base controls; (ii) firm fixed effects with base controls; (iii)–(iv) the same two specifications with

extended controls; and (v)–(vi) the two extended-control specifications with calendar year-quarter fixed effects added. The base control set follows [Bates et al. \(2009\)](#) (firm size, leverage, market-to-book, ROA, capital expenditure, dividend payer, and operating-cash-flow volatility); the extended set adds sales growth, R&D intensity, cash flow, and stock-return volatility. The lagged dependent variable is included as a base control in all specifications to absorb cash-policy persistence.

Standard errors are clustered at the firm level. Tests of the main hypothesis are one-tailed in the predicted direction; tests of the amplification hypotheses are one-tailed in the predicted direction of the interaction sign.

## II.5 Specification and Measurement of Key Constructs

Before turning to the cash analyses, we evaluate the construct validity of the speech-uncertainty measure by testing whether it covaries with established external proxies for firm-relevant uncertainty. If the measure captures uncertainty information rather than mechanical text-frequency artifacts, it should move with proxies that have been validated as uncertainty signals in prior literature.

We use four external proxies: firm-level political-risk exposure measured from earnings-call discussion of political topics ([Hassan, Hollander, van Lent, & Tahoun, 2020](#)); aggregate U.S. economic policy uncertainty constructed from newspaper coverage ([Baker, Bloom, & Davis, 2016](#)); global economic policy uncertainty as a GDP-weighted average across countries ([Davis, 2016](#)); and product-market competition intensity from text-based industry-network similarity ([Hoberg & Phillips, 2016](#)).

We expect UncResCEO and UncPreCEO—the within-CEO time-varying components of the speech-uncertainty measure—to load positively on the four proxies. ClarityCEO is a CEO-level fixed effect and is excluded from this test. The construct-validity results are reported alongside the main empirical analyses.

## III Main Empirical Analyses

### III.A Data, Sample, and Variable Construction

Our sample covers earnings-call observations of U.S. public firms over the period 2002 through 2018 at the firm-quarter frequency. The full call panel comprises

112,968 firm-quarter observations across 2,429 distinct firms. We obtain financial variables from Compustat and earnings-call text features from a vendor-supplied transcript repository.

We exclude firms in financial services and utilities (Fama-French 12 industries 11 and 8). The estimation sample also requires a non-missing lagged dependent variable and is restricted to firms with multiple observations after firm-level fixed-effect singleton dropping. The resulting sample comprises approximately 43,300 firm-quarter observations across 1,376 firms in the contemporaneous specifications and approximately 41,100 firm-quarter observations across 1,321 firms in the lead specifications.

Section II describes the construction of the three CEO speech-uncertainty independent variables (ClarityCEO, UncResCEO, UncPreCEO). The cash-to-assets ratio ( $\text{Cash}_t, \text{Cash}_{t+1}$ ) is cash and short-term investments scaled by total assets. Variable definitions are provided in the appendix; summary statistics are reported in Table 1.

### III.B Cash Holdings (HC)

Table 2 reports the main estimation results for cash holdings on the three CEO speech-uncertainty independent variables. Across the twelve specifications, UncResCEO loads positively on cash holdings in every cell, with one-tailed p-values below the ten-percent level.

ClarityCEO loads negatively in all twelve specifications, consistent with the precautionary direction (lower clarity, higher cash); the loading is significant at the ten-percent level under one-tailed tests in nine of twelve cells. UncPreCEO is positive but statistically indistinguishable from zero across all twelve specifications.

As a robustness check, Table 3 replicates the main results using a quarterly-expanding-window construction of ClarityCEO and UncResCEO that eliminates look-ahead bias by re-estimating the speech-uncertainty decomposition each quarter from data available through that quarter only. UncResCEO remains positive and significant in twelve of twelve specifications; the ClarityCEO and UncPreCEO patterns are preserved.

### III.C Constraint Amplification (HFC)

H1a predicts that the cash response amplifies under financing-friction stress, with constrained firms (proxied by an unrated debt status) showing a stronger

relation between speech uncertainty and cash holdings than rated firms (Almeida et al., 2004). Table 4 reports estimates of the H1 specification with the addition of an Unrated dummy and its interactions with UncResCEO and UncPreCEO. ClarityCEO is not interacted with Unrated, since the constructed CEO trait does not separate cleanly from cross-sectional firm constraints.

The UncResCEO  $\times$  Unrated interaction loads positively across all eight interaction specifications. The interaction is significant at the ten-percent level under one-tailed tests in the lead-cash specifications with industry fixed effects (columns 13 and 15) but is not significant in the contemporaneous specifications or in the firm-fixed-effects cells. The UncPreCEO  $\times$  Unrated interaction is direction-consistent in all eight specifications and is significant in one cell (column 7).

The amplification pattern is direction-consistent in every cell. Financing friction amplifies the cash response in the cross-section, with statistical detection in the lead-cash specifications under industry fixed effects. The within-firm variation in the Unrated dummy is sparse—firms rarely transition between rated and unrated status within the sample period—so the interaction coefficient is identified primarily off cross-firm variation in financing friction. Table 5 replicates the analysis using the quarterly-expanding-window construction; the pattern is preserved.

### III.D Cash-Flow-Volatility Amplification

H1b predicts that the cash response amplifies under cash-flow-volatility stress, with high-volatility firms showing a stronger relation between speech uncertainty and cash holdings than low-volatility firms (Han & Qiu, 2007). Cash-flow volatility (CFvol) is constructed as the coefficient of variation of operating cash flow over a sixteen-quarter rolling window, following Han and Qiu (2007). The HighCFvol indicator equals one when a firm-quarter is at or above the Fama-French 12 industry-year median CFvol. Table 6 reports estimates of the H1 specification with the addition of HighCFvol and its interactions with UncResCEO and UncPreCEO.

The UncPreCEO  $\times$  HighCFvol interaction loads positively in all eight specifications and is significant at the ten-percent level under one-tailed tests in four of eight cells—all four with industry fixed effects, across both the contemporaneous and lead horizons. The UncResCEO  $\times$  HighCFvol interaction is direction-consistent in all eight cells and significant in two

of eight (lead-cash specifications with industry fixed effects). The interaction effects are absorbed under firm fixed effects, mirroring the pattern in Section III.

Cash-flow-volatility stress amplifies the cash response in the cross-section, with statistical detection across both horizons under industry fixed effects. The H1b specification adopts the cash-flow-volatility construction of Han and Qiu (2007) as a moderator on the speech–cash relation. Together with the financing-friction amplification of Section III, the two interactions cover both stress dimensions of the precautionary cash motive.

### III.E Endogeneity

The most consequential endogeneity concern in our setting is reverse causality: cash policy could shape CEO speech rather than the other way around. Within-firm fixed effects and lagged-DV controls absorb time-invariant firm heterogeneity and policy persistence, but neither rules out the reverse arrow at the within-firm time-series level. We address this concern through cross-sectional *modus tollens* on the two precautionary triggers identified in the conceptual framework: if the forward direction (speech uncertainty  $\rightarrow$  cash) operates through the precautionary mechanism, the cash response should amplify under stress conditions that are themselves cross-sectionally exogenous to within-firm cash-policy choices.

Sections III and III provide such evidence. The cash response amplifies in the cross-section under both financing-friction stress (rated versus unrated debt status) and cash-flow-volatility stress (high versus low CFvol). Neither moderator is plausibly endogenous to within-firm CEO speech: rated/unrated status is determined primarily by firm size and reputation external to quarterly speech variation, and CFvol is constructed from a sixteen-quarter trailing window, predating the contemporaneous speech measure. The observed amplification pattern is consistent with the forward direction and harder to reconcile with a reverse-causation alternative.

We acknowledge as a limitation that no exogenous-shock or instrumental-variable identification is pursued. We considered and dropped six candidate designs during the project: a CEO sudden-death difference-in-differences, a first-difference reformulation of the Dzielinski-Wagner-Zeckhauser specification, a Lewbel heteroskedasticity-based instrument, weather-shock instruments, the American Jobs Creation Act repatriation experiment, and the Almeida-Campello-Laranjeira-Weisbenner long-



term-debt-maturity difference-in-differences. Each failed for design-specific reasons including mechanism mismatch, parallel-trends violations, dual-channel contamination, and absence of a documented cash response to the candidate shock.

The most promising remaining candidate is the firm-level political-risk text measure of Hassan et al. (2020) interacted with major political shocks. We flag this direction as future work but do not pursue it in the present paper.

## IV Additional Analyses

### IV.A Market Information-Asymmetry Channel: Post-Call Bid-Ask Spread

[§IV.A body pending Step 6 prose population.]

### IV.B Disclosure-Insufficiency Channel: Conference-Call Comment Letter (CCCL)

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## V Conclusion

### V.1 Summary

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### V.2 Limitations

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### V.3 Future Work

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Table 1: Summary Statistics

| Variable                        | Unit | N      | Mean    | SD       | Min       | P25     | Median  | P75      | Max       |
|---------------------------------|------|--------|---------|----------|-----------|---------|---------|----------|-----------|
| <i>A. Independent Variables</i> |      |        |         |          |           |         |         |          |           |
| ClarityCEO                      | C    | 61,626 | 0.2143  | 0.2129   | -0.7717   | 0.0952  | 0.2365  | 0.3629   | 0.8614    |
| ClarityCEO_QtrExp               | C    | 25,520 | 0.2011  | 0.2281   | -1.0232   | 0.0713  | 0.2256  | 0.3589   | 0.8694    |
| GEPU_log                        | C    | 88,205 | 4.6872  | 0.3672   | 3.8626    | 4.3797  | 4.7238  | 4.9666   | 5.5106    |
| PRisk                           | C    | 85,995 | 99.6187 | 146.5189 | 0.0000    | 17.0778 | 53.3729 | 120.1973 | 1192.4308 |
| PRisk_lag2                      | C    | 80,627 | 99.9614 | 147.2392 | 0.0000    | 17.1702 | 53.4446 | 120.2666 | 1192.4308 |
| TotalSimilarity                 | F    | 22,403 | 3.4277  | 5.6522   | 1.0000    | 1.1246  | 1.5865  | 3.0903   | 80.1876   |
| US_EPU_log                      | C    | 88,205 | 4.7123  | 0.3661   | 3.8018    | 4.4805  | 4.6891  | 4.9625   | 5.6478    |
| UncPreCEO                       | C    | 71,727 | 0.6602  | 0.3917   | 0.0000    | 0.3810  | 0.6001  | 0.8723   | 2.2432    |
| UncResCEO                       | C    | 44,900 | -0.0000 | 0.3010   | -1.5108   | -0.1846 | -0.0180 | 0.1650   | 1.7794    |
| UncResCEO_QtrExp                | C    | 25,520 | -0.0057 | 0.3208   | -1.6084   | -0.2028 | -0.0210 | 0.1707   | 1.7402    |
| Unrated                         | C    | 79,487 | 0.5159  | 0.4997   | 0.0000    | 0.0000  | 1.0000  | 1.0000   | 1.0000    |
| <i>B. Dependent Variables</i>   |      |        |         |          |           |         |         |          |           |
| BGTLevel_Spread                 | C    | 87,018 | 0.0017  | 0.0026   | 0.0001    | 0.0004  | 0.0009  | 0.0017   | 0.0164    |
| BGTLevel_Spread_lead1           | C    | 82,843 | 0.0016  | 0.0024   | 0.0001    | 0.0004  | 0.0009  | 0.0016   | 0.0164    |
| CCCL                            | C    | 88,205 | 0.0032  | 0.0563   | 0.0000    | 0.0000  | 0.0000  | 0.0000   | 1.0000    |
| CashRatio                       | C    | 87,990 | 0.1708  | 0.1806   | 0.0000    | 0.0368  | 0.1047  | 0.2435   | 0.9938    |
| CashRatio_lead                  | C    | 82,236 | 0.1715  | 0.1748   | 0.0000    | 0.0407  | 0.1112  | 0.2447   | 0.9938    |
| <i>C. Firm Controls</i>         |      |        |         |          |           |         |         |          |           |
| AbsSurpDec                      | C    | 61,544 | 2.9300  | 1.4255   | 0.0000    | 2.0000  | 3.0000  | 4.0000   | 5.0000    |
| Capex                           | C    | 87,627 | 0.0487  | 0.0536   | 0.0000    | 0.0176  | 0.0324  | 0.0592   | 0.5203    |
| CashFlowAt                      | C    | 87,670 | 0.1019  | 0.1147   | -3.0324   | 0.0608  | 0.1032  | 0.1526   | 0.4874    |
| DailyVola                       | C    | 82,599 | 36.7556 | 21.2437  | 9.4388    | 22.9709 | 31.2762 | 43.8148  | 211.9160  |
| DivDummy                        | C    | 88,134 | 0.4755  | 0.4994   | 0.0000    | 0.0000  | 0.0000  | 1.0000   | 1.0000    |
| EarnVol                         | C    | 87,440 | 0.0701  | 0.1907   | 0.0006    | 0.0163  | 0.0315  | 0.0695   | 19.5101   |
| FirmMat                         | C    | 87,571 | -0.0432 | 2.1826   | -317.5714 | -0.0078 | 0.2191  | 0.4195   | 0.9252    |
| Leverage                        | C    | 87,994 | 0.2330  | 0.2240   | 0.0000    | 0.0470  | 0.2065  | 0.3455   | 3.9453    |
| NegCall                         | C    | 86,858 | 0.9210  | 0.3024   | 0.0000    | 0.7038  | 0.8798  | 1.0943   | 2.3901    |
| RDSales                         | C    | 87,647 | 0.1047  | 0.9342   | 0.0000    | 0.0000  | 0.0048  | 0.0636   | 39.7688   |
| ROA                             | C    | 87,552 | 0.0386  | 0.1330   | -3.9826   | 0.0146  | 0.0525  | 0.0925   | 0.5490    |
| SalesGrowth                     | C    | 87,506 | 0.1173  | 0.3849   | -1.0000   | -0.0043 | 0.0745  | 0.1762   | 10.3051   |
| StockPrice                      | C    | 87,188 | 36.8864 | 32.8119  | 1.6100    | 14.8975 | 28.2400 | 48.1700  | 192.0384  |
| TobinsQ                         | C    | 87,363 | 2.1363  | 1.5197   | 0.3847    | 1.2681  | 1.6791  | 2.4362   | 35.6144   |
| Turnover                        | C    | 87,188 | 0.0260  | 0.0299   | 0.0007    | 0.0083  | 0.0160  | 0.0310   | 0.1697    |
| UncQue                          | C    | 81,031 | 1.4398  | 0.4831   | 0.0000    | 1.1216  | 1.4108  | 1.7268   | 3.5945    |
| lnAssets                        | C    | 87,994 | 7.4710  | 1.6607   | -0.5692   | 6.3008  | 7.3763  | 8.5436   | 12.4592   |
| sCFO                            | C    | 83,662 | 0.0619  | 0.2233   | 0.0014    | 0.0236  | 0.0395  | 0.0664   | 32.4035   |

*Notes:* Summary statistics for variables used in the hypothesis tests. Sample period: 2002–2018. Anchor sample (H1, Main): 88,205 earnings-call observations across 1,884 firms. Unit column: C = call-level, F = firm-year. N varies by row because each variable is summarized on its anchor panel's Main sample (complete cases per variable); per-suite samples differ due to lags, control availability, and merges.

Table 2: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO

|                         | (1)                           | (2)                           | (3)                           | (4)                           | (5)                           | (6)                           | (7)                           | (8)                           | (9)                           | (10)                          | (11)                          | (12)                          |
|-------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                         | CashRatio                     |                               |                               |                               |                               |                               | CashRatio_lead                |                               |                               |                               |                               |                               |
| CEO Clarity (DWZ)       | <b>-0.0046*</b><br>(0.0033)   | <b>-0.0071*</b><br>(0.0051)   | <b>-0.0058**</b><br>(0.0033)  | -0.0054<br>(0.0048)           | <b>-0.0056**</b><br>(0.0033)  | -0.0052<br>(0.0047)           | -0.0052<br>(0.0052)           | <b>-0.0170**</b><br>(0.0095)  | <b>-0.0068*</b><br>(0.0051)   | <b>-0.0162**</b><br>(0.0090)  | <b>-0.0069*</b><br>(0.0051)   | <b>-0.0161**</b><br>(0.0090)  |
| CEO Residual Unc. (DWZ) | <b>0.0021**</b><br>(0.0010)   | <b>0.0019**</b><br>(0.0010)   | <b>0.0020**</b><br>(0.0010)   | <b>0.0016*</b><br>(0.0010)    | <b>0.0021**</b><br>(0.0010)   | <b>0.0016*</b><br>(0.0010)    | <b>0.0028**</b><br>(0.0014)   | <b>0.0023**</b><br>(0.0013)   | <b>0.0026**</b><br>(0.0014)   | <b>0.0020*</b><br>(0.0013)    | <b>0.0025**</b><br>(0.0014)   | <b>0.0019*</b><br>(0.0013)    |
| CEO Pres Uncertainty    | 0.0002<br>(0.0011)            | 0.0009<br>(0.0011)            | 0.0003<br>(0.0011)            | 0.0006<br>(0.0011)            | 0.0004<br>(0.0011)            | 0.0007<br>(0.0011)            | -0.0003<br>(0.0021)           | 0.0008<br>(0.0017)            | -0.0005<br>(0.0021)           | 0.0004<br>(0.0017)            | -0.0007<br>(0.0021)           | 0.0002<br>(0.0017)            |
| Leverage                | <b>-0.0177***</b><br>(0.0048) | <b>-0.0355***</b><br>(0.0084) | <b>-0.0130***</b><br>(0.0036) | <b>-0.0218***</b><br>(0.0082) | <b>-0.0132***</b><br>(0.0037) | <b>-0.0216***</b><br>(0.0082) | <b>-0.0327***</b><br>(0.0120) | <b>-0.0279**</b><br>(0.0121)  | <b>-0.0274***</b><br>(0.0104) | -0.0159<br>(0.0118)           | <b>-0.0275***</b><br>(0.0105) | -0.0156<br>(0.0118)           |
| lnAssets                | <b>-0.0015***</b><br>(0.0006) | <b>-0.0177***</b><br>(0.0026) | <b>-0.0017***</b><br>(0.0005) | <b>-0.0175***</b><br>(0.0026) | <b>-0.0014***</b><br>(0.0005) | <b>-0.0170***</b><br>(0.0026) | <b>-0.0024**</b><br>(0.0010)  | <b>-0.0268***</b><br>(0.0042) | <b>-0.0020**</b><br>(0.0009)  | <b>-0.0265***</b><br>(0.0042) | <b>-0.0020**</b><br>(0.0010)  | <b>-0.0265***</b><br>(0.0042) |
| TobinsQ                 | <b>0.0058***</b><br>(0.0006)  | <b>0.0048***</b><br>(0.0009)  | <b>0.0042***</b><br>(0.0007)  | <b>0.0037***</b><br>(0.0010)  | <b>0.0040***</b><br>(0.0007)  | <b>0.0034***</b><br>(0.0010)  | <b>0.0085***</b><br>(0.0012)  | 0.0025<br>(0.0015)            | <b>0.0056***</b><br>(0.0012)  | 0.0009<br>(0.0015)            | <b>0.0056***</b><br>(0.0012)  | 0.0010<br>(0.0016)            |
| ROA                     | -0.0144<br>(0.0093)           | <b>0.0214**</b><br>(0.0105)   | <b>-0.0679***</b><br>(0.0113) | -0.0195<br>(0.0122)           | <b>-0.0630***</b><br>(0.0114) | -0.0171<br>(0.0122)           | <b>-0.0762***</b><br>(0.0161) | <b>-0.0352*</b><br>(0.0185)   | <b>-0.1427***</b><br>(0.0173) | <b>-0.0790***</b><br>(0.0194) | <b>-0.1378***</b><br>(0.0174) | <b>-0.0769***</b><br>(0.0196) |
| Capex                   | <b>-0.1117***</b><br>(0.0128) | <b>-0.2995***</b><br>(0.0234) | <b>-0.1596***</b><br>(0.0149) | <b>-0.3074***</b><br>(0.0236) | <b>-0.1564***</b><br>(0.0148) | <b>-0.2993***</b><br>(0.0234) | <b>-0.1613***</b><br>(0.0184) | <b>-0.3281***</b><br>(0.0344) | <b>-0.2433***</b><br>(0.0216) | <b>-0.3415***</b><br>(0.0345) | <b>-0.2411***</b><br>(0.0216) | <b>-0.3430***</b><br>(0.0346) |
| DivDummy                | <b>-0.0041***</b><br>(0.0013) | -0.0028<br>(0.0023)           | <b>-0.0054***</b><br>(0.0013) | <b>-0.0041*</b><br>(0.0022)   | <b>-0.0049***</b><br>(0.0013) | -0.0036<br>(0.0022)           | <b>-0.0051**</b><br>(0.0023)  | -0.0068<br>(0.0045)           | <b>-0.0056**</b><br>(0.0023)  | <b>-0.0076*</b><br>(0.0044)   | <b>-0.0054**</b><br>(0.0023)  | <b>-0.0077*</b><br>(0.0044)   |
| sCFO                    | <b>0.0066***</b><br>(0.0020)  | -0.0022<br>(0.0014)           | <b>0.0079***</b><br>(0.0030)  | -0.0008<br>(0.0014)           | <b>0.0074***</b><br>(0.0028)  | -0.0009<br>(0.0014)           | <b>0.0137**</b><br>(0.0057)   | 0.0027<br>(0.0025)            | <b>0.0137**</b><br>(0.0062)   | 0.0035<br>(0.0027)            | <b>0.0134**</b><br>(0.0061)   | 0.0034<br>(0.0027)            |
| Lagged_DV               | <b>0.8558***</b><br>(0.0076)  | <b>0.6146***</b><br>(0.0133)  | <b>0.8612***</b><br>(0.0075)  | <b>0.6322***</b><br>(0.0130)  | <b>0.8617***</b><br>(0.0075)  | <b>0.6337***</b><br>(0.0130)  | <b>0.7214***</b><br>(0.0138)  | <b>0.2209***</b><br>(0.0201)  | <b>0.7227***</b><br>(0.0133)  | <b>0.2377***</b><br>(0.0201)  | <b>0.7231***</b><br>(0.0133)  | <b>0.2385***</b><br>(0.0201)  |
| SalesGrowth             |                               |                               | <b>-0.0178***</b><br>(0.0045) | <b>-0.0274***</b><br>(0.0052) | <b>-0.0169***</b><br>(0.0044) | <b>-0.0266***</b><br>(0.0052) |                               |                               | <b>-0.0156**</b><br>(0.0061)  | <b>-0.0171***</b><br>(0.0050) | <b>-0.0144**</b><br>(0.0060)  | <b>-0.0163***</b><br>(0.0050) |
| RDSales                 |                               |                               | <b>0.0062***</b><br>(0.0017)  | <b>-0.0032***</b><br>(0.0012) | <b>0.0062***</b><br>(0.0017)  | <b>-0.0031**</b><br>(0.0012)  |                               |                               | <b>0.0137***</b><br>(0.0024)  | -0.0014<br>(0.0019)           | <b>0.0138***</b><br>(0.0024)  | -0.0012<br>(0.0019)           |
| CashFlowAt              |                               |                               | <b>0.1292***</b><br>(0.0162)  | <b>0.1589***</b><br>(0.0173)  | <b>0.1278***</b><br>(0.0162)  | <b>0.1576***</b><br>(0.0173)  |                               |                               | <b>0.1982***</b><br>(0.0236)  | <b>0.1687***</b><br>(0.0249)  | <b>0.1944***</b><br>(0.0237)  | <b>0.1660***</b><br>(0.0250)  |
| DailyVola               |                               |                               | 0.0000<br>(0.0000)            | <b>-0.0001***</b><br>(0.0000) | <b>0.0001***</b><br>(0.0000)  | <b>-0.0001*</b><br>(0.0000)   |                               |                               | <b>0.0003***</b><br>(0.0000)  | <b>0.0001***</b><br>(0.0000)  | <b>0.0003***</b><br>(0.0001)  | 0.0001<br>(0.0001)            |
| Extended Controls       |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |
| Industry FE             | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               |
| Firm FE                 |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |
| Year FE                 | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               |
| Year-Quarter FE         |                               |                               |                               |                               | Yes                           | Yes                           |                               |                               |                               |                               | Yes                           | Yes                           |
| N                       | 43,333                        | 43,333                        | 43,316                        | 43,316                        | 43,316                        | 43,316                        | 41,122                        | 41,122                        | 41,108                        | 41,108                        | 41,108                        | 41,108                        |
| R <sup>2</sup>          | 0.825                         | 0.431                         | 0.829                         | 0.449                         | 0.830                         | 0.449                         | 0.659                         | 0.092                         | 0.667                         | 0.108                         | 0.668                         | 0.108                         |
| Adj. R <sup>2</sup>     | 0.825                         | 0.412                         | 0.829                         | 0.430                         | 0.830                         | 0.430                         | 0.658                         | 0.062                         | 0.667                         | 0.078                         | 0.667                         | 0.077                         |

Notes: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms).  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).



Table 3: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO

|                                 | (1)                           | (2)                           | (3)                           | (4)                           | (5)                           | (6)                           | (7)                           | (8)                           | (9)                           | (10)                          | (11)                          | (12)                          |
|---------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                                 | CashRatio                     |                               |                               |                               |                               |                               | CashRatio_Lead                |                               |                               |                               |                               |                               |
| CEO Clarity (DWZ Qtr-Exp)       | <b>-0.0083***</b><br>(0.0033) | <b>-0.0148***</b><br>(0.0047) | <b>-0.0093***</b><br>(0.0033) | <b>-0.0129***</b><br>(0.0045) | <b>-0.0092***</b><br>(0.0033) | <b>-0.0126***</b><br>(0.0045) | -0.0024<br>(0.0053)           | <b>-0.0215**</b><br>(0.0097)  | -0.0041<br>(0.0053)           | <b>-0.0210**</b><br>(0.0094)  | -0.0042<br>(0.0053)           | <b>-0.0210**</b><br>(0.0095)  |
| CEO Residual Unc. (DWZ Qtr-Exp) | <b>0.0018*</b><br>(0.0012)    | <b>0.0020**</b><br>(0.0012)   | <b>0.0016*</b><br>(0.0012)    | <b>0.0018*</b><br>(0.0012)    | <b>0.0016*</b><br>(0.0012)    | <b>0.0017*</b><br>(0.0012)    | <b>0.0046***</b><br>(0.0019)  | <b>0.0036**</b><br>(0.0017)   | <b>0.0044**</b><br>(0.0019)   | <b>0.0033**</b><br>(0.0017)   | <b>0.0044**</b><br>(0.0019)   | <b>0.0033**</b><br>(0.0017)   |
| CEO Pres Uncertainty            | 0.0001<br>(0.0013)            | 0.0006<br>(0.0013)            | -0.0001<br>(0.0013)           | 0.0000<br>(0.0013)            | -0.0001<br>(0.0013)           | 0.0001<br>(0.0013)            | 0.0002<br>(0.0024)            | 0.0016<br>(0.0021)            | -0.0003<br>(0.0024)           | 0.0011<br>(0.0021)            | -0.0004<br>(0.0024)           | 0.0011<br>(0.0021)            |
| Leverage                        | <b>-0.0131***</b><br>(0.0040) | <b>-0.0324***</b><br>(0.0086) | <b>-0.0089***</b><br>(0.0031) | <b>-0.0199**</b><br>(0.0085)  | <b>-0.0087***</b><br>(0.0031) | <b>-0.0190**</b><br>(0.0085)  | <b>-0.0293**</b><br>(0.0122)  | <b>-0.0295**</b><br>(0.0139)  | <b>-0.0241**</b><br>(0.0108)  | -0.0181<br>(0.0136)           | <b>-0.0244**</b><br>(0.0109)  | -0.0179<br>(0.0136)           |
| lnAssets                        | -0.0009<br>(0.0006)           | <b>-0.0169***</b><br>(0.0028) | <b>-0.0011*</b><br>(0.0006)   | <b>-0.0169***</b><br>(0.0027) | <b>-0.0011*</b><br>(0.0006)   | <b>-0.0167***</b><br>(0.0027) | <b>-0.0018*</b><br>(0.0011)   | <b>-0.0285***</b><br>(0.0049) | -0.0016<br>(0.0010)           | <b>-0.0286***</b><br>(0.0049) | -0.0015<br>(0.0011)           | <b>-0.0287***</b><br>(0.0050) |
| TobinsQ                         | <b>0.0052***</b><br>(0.0007)  | <b>0.0039***</b><br>(0.0011)  | <b>0.0034***</b><br>(0.0009)  | <b>0.0028**</b><br>(0.0012)   | <b>0.0033***</b><br>(0.0009)  | <b>0.0025**</b><br>(0.0012)   | <b>0.0079***</b><br>(0.0012)  | 0.0011<br>(0.0017)            | <b>0.0046***</b><br>(0.0014)  | -0.0009<br>(0.0017)           | <b>0.0046***</b><br>(0.0014)  | -0.0009<br>(0.0018)           |
| ROA                             | -0.0107<br>(0.0109)           | <b>0.0230*</b><br>(0.0124)    | <b>-0.0510***</b><br>(0.0131) | -0.0135<br>(0.0144)           | <b>-0.0498***</b><br>(0.0132) | -0.0129<br>(0.0144)           | <b>-0.0621***</b><br>(0.0192) | -0.0188<br>(0.0227)           | <b>-0.1280***</b><br>(0.0202) | <b>-0.0721***</b><br>(0.0237) | <b>-0.1270***</b><br>(0.0204) | <b>-0.0731***</b><br>(0.0240) |
| Capex                           | <b>-0.0967***</b><br>(0.0150) | <b>-0.2472***</b><br>(0.0253) | <b>-0.1415***</b><br>(0.0169) | <b>-0.2628***</b><br>(0.0258) | <b>-0.1414***</b><br>(0.0168) | <b>-0.2605***</b><br>(0.0257) | <b>-0.1613***</b><br>(0.0214) | <b>-0.2816***</b><br>(0.0372) | <b>-0.2420***</b><br>(0.0249) | <b>-0.3002***</b><br>(0.0375) | <b>-0.2423***</b><br>(0.0249) | <b>-0.2996***</b><br>(0.0375) |
| DivDummy                        | <b>-0.0037**</b><br>(0.0015)  | -0.0021<br>(0.0026)           | <b>-0.0051***</b><br>(0.0015) | -0.0034<br>(0.0025)           | <b>-0.0050***</b><br>(0.0015) | -0.0033<br>(0.0025)           | <b>-0.0042*</b><br>(0.0025)   | -0.0080<br>(0.0054)           | <b>-0.0048**</b><br>(0.0025)  | <b>-0.0090*</b><br>(0.0053)   | <b>-0.0047*</b><br>(0.0025)   | <b>-0.0091*</b><br>(0.0053)   |
| sCFO                            | <b>0.0090***</b><br>(0.0024)  | -0.0008<br>(0.0017)           | <b>0.0100***</b><br>(0.0031)  | 0.0004<br>(0.0014)            | <b>0.0099***</b><br>(0.0030)  | 0.0004<br>(0.0015)            | <b>0.0147*</b><br>(0.0081)    | 0.0031<br>(0.0032)            | <b>0.0143*</b><br>(0.0082)    | 0.0038<br>(0.0032)            | <b>0.0142*</b><br>(0.0081)    | 0.0037<br>(0.0032)            |
| Lagged_DV                       | <b>0.8718***</b><br>(0.0091)  | <b>0.6433***</b><br>(0.0159)  | <b>0.8734***</b><br>(0.0091)  | <b>0.6576***</b><br>(0.0156)  | <b>0.8738***</b><br>(0.0091)  | <b>0.6579***</b><br>(0.0156)  | <b>0.7424***</b><br>(0.0150)  | <b>0.2158***</b><br>(0.0242)  | <b>0.7404***</b><br>(0.0148)  | <b>0.2320***</b><br>(0.0242)  | <b>0.7402***</b><br>(0.0148)  | <b>0.2317***</b><br>(0.0242)  |
| SalesGrowth                     |                               |                               | <b>-0.0165***</b><br>(0.0054) | <b>-0.0257***</b><br>(0.0068) | <b>-0.0165***</b><br>(0.0054) | <b>-0.0256***</b><br>(0.0068) |                               |                               | -0.0106<br>(0.0070)           | <b>-0.0137**</b><br>(0.0064)  | -0.0107<br>(0.0070)           | <b>-0.0136**</b><br>(0.0064)  |
| RDSales                         |                               |                               | <b>0.0091***</b><br>(0.0025)  | -0.0004<br>(0.0025)           | <b>0.0091***</b><br>(0.0025)  | -0.0003<br>(0.0017)           |                               |                               | <b>0.0160***</b><br>(0.0038)  | 0.0005<br>(0.0021)            | <b>0.0159***</b><br>(0.0038)  | 0.0005<br>(0.0021)            |
| CashFlowAt                      |                               |                               | <b>0.1182***</b><br>(0.0159)  | <b>0.1536***</b><br>(0.0175)  | <b>0.1189***</b><br>(0.0159)  | <b>0.1544***</b><br>(0.0175)  |                               |                               | <b>0.1977***</b><br>(0.0275)  | <b>0.1883***</b><br>(0.0300)  | <b>0.1974***</b><br>(0.0275)  | <b>0.1880***</b><br>(0.0300)  |
| DailyVola                       |                               |                               |                               | 0.0000<br>(0.0000)            | <b>-0.0001**</b><br>(0.0000)  | <b>-0.0001**</b><br>(0.0000)  |                               |                               | <b>0.0002***</b><br>(0.0001)  | 0.0000<br>(0.0000)            | <b>0.0003***</b><br>(0.0001)  | 0.0000<br>(0.0001)            |
| Extended Controls               |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |
| Industry FE                     | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               |
| Firm FE                         |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |
| Year FE                         | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               |
| Year-Quarter FE                 |                               |                               |                               |                               | Yes                           | Yes                           |                               |                               |                               |                               | Yes                           | Yes                           |
| N                               | 24,868                        | 24,868                        | 24,855                        | 24,855                        | 24,855                        | 24,855                        | 23,127                        | 23,127                        | 23,120                        | 23,120                        | 23,120                        | 23,120                        |
| R <sup>2</sup>                  | 0.839                         | 0.457                         | 0.842                         | 0.473                         | 0.842                         | 0.473                         | 0.675                         | 0.087                         | 0.683                         | 0.105                         | 0.683                         | 0.105                         |
| Adj. R <sup>2</sup>             | 0.839                         | 0.430                         | 0.841                         | 0.446                         | 0.842                         | 0.446                         | 0.674                         | 0.041                         | 0.682                         | 0.059                         | 0.682                         | 0.058                         |

Notes: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms).  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 4: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO  $\times$  Unrated Constraint

|                            | (1)                           | (2)                           | (3)                           | (4)                           | (5)                           | (6)                           | (7)                           | (8)                           |
|----------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                            |                               | CashRatio                     |                               |                               |                               | CashRatio <sub>lead</sub>     |                               |                               |
| CEO Clarity (DWZ, c)       | <b>-0.0062**</b><br>(0.0034)  | -0.0043<br>(0.0052)           | <b>-0.0059**</b><br>(0.0034)  | -0.0042<br>(0.0052)           | -0.0052<br>(0.0053)           | <b>-0.0157*</b><br>(0.0096)   | -0.0052<br>(0.0053)           | <b>-0.0155*</b><br>(0.0096)   |
| CEO Residual Unc. (DWZ, c) | <b>0.0020**</b><br>(0.0011)   | <b>0.0018**</b><br>(0.0010)   | <b>0.0022**</b><br>(0.0011)   | <b>0.0019**</b><br>(0.0010)   | <b>0.0027**</b><br>(0.0015)   | <b>0.0022*</b><br>(0.0013)    | <b>0.0026**</b><br>(0.0015)   | <b>0.0021*</b><br>(0.0013)    |
| CEO Pres Unc. (c)          | 0.0002<br>(0.0012)            | 0.0005<br>(0.0012)            | 0.0003<br>(0.0012)            | 0.0007<br>(0.0012)            | -0.0007<br>(0.0022)           | -0.0001<br>(0.0018)           | -0.0008<br>(0.0022)           | -0.0002<br>(0.0018)           |
| Unrated                    | <b>0.0038***</b><br>(0.0012)  | -0.0027<br>(0.0035)           | -0.0010<br>(0.0016)           | -0.0026<br>(0.0035)           | <b>0.0065***</b><br>(0.0022)  | <b>-0.0093*</b><br>(0.0054)   | -0.0014<br>(0.0029)           | <b>-0.0095*</b><br>(0.0054)   |
| UncRes $\times$ Unrated    | 0.0011<br>(0.0021)            | -0.0001<br>(0.0020)           | 0.0012<br>(0.0021)            | -0.0000<br>(0.0020)           | <b>0.0057**</b><br>(0.0029)   | 0.0033<br>(0.0026)            | <b>0.0057**</b><br>(0.0029)   | 0.0033<br>(0.0026)            |
| UncPre $\times$ Unrated    | 0.0024<br>(0.0022)            | 0.0013<br>(0.0025)            | <b>0.0029*</b><br>(0.0025)    | 0.0014<br>(0.0025)            | 0.0005<br>(0.0040)            | -0.0026<br>(0.0038)           | 0.0010<br>(0.0040)            | -0.0027<br>(0.0038)           |
| Leverage                   | <b>-0.0117***</b><br>(0.0033) | <b>-0.0216**</b><br>(0.0098)  | <b>-0.0141***</b><br>(0.0041) | <b>-0.0214**</b><br>(0.0099)  | <b>-0.0237**</b><br>(0.0103)  | -0.0107<br>(0.0131)           | <b>-0.0274**</b><br>(0.0117)  | -0.0104<br>(0.0131)           |
| lnAssets                   | <b>0.0005*</b><br>(0.0003)    | <b>-0.0159***</b><br>(0.0027) | <b>-0.0019***</b><br>(0.0007) | <b>-0.0155***</b><br>(0.0027) | <b>0.0018***</b><br>(0.0005)  | <b>-0.0273***</b><br>(0.0044) | <b>-0.0025**</b><br>(0.0011)  | <b>-0.0275***</b><br>(0.0044) |
| TobinsQ                    | <b>0.0047***</b><br>(0.0008)  | <b>0.0045***</b><br>(0.0011)  | <b>0.0041***</b><br>(0.0008)  | <b>0.0040***</b><br>(0.0011)  | <b>0.0059***</b><br>(0.0013)  | 0.0009<br>(0.0017)            | <b>0.0052***</b><br>(0.0013)  | 0.0010<br>(0.0017)            |
| ROA                        | <b>-0.0715***</b><br>(0.0122) | -0.0157<br>(0.0122)           | <b>-0.0651***</b><br>(0.0122) | -0.0133<br>(0.0122)           | <b>-0.1452***</b><br>(0.0182) | <b>-0.0819***</b><br>(0.0200) | <b>-0.1378***</b><br>(0.0183) | <b>-0.0800***</b><br>(0.0202) |
| Capex                      | <b>-0.1610***</b><br>(0.0158) | <b>-0.3087***</b><br>(0.0254) | <b>-0.1647***</b><br>(0.0158) | <b>-0.3005***</b><br>(0.0251) | <b>-0.2338***</b><br>(0.0226) | <b>-0.3399***</b><br>(0.0360) | <b>-0.2426***</b><br>(0.0227) | <b>-0.3419***</b><br>(0.0361) |
| DivDummy                   | <b>-0.0058***</b><br>(0.0013) | <b>-0.0043*</b><br>(0.0023)   | <b>-0.0054***</b><br>(0.0013) | -0.0038<br>(0.0023)           | <b>-0.0054**</b><br>(0.0023)  | -0.0065<br>(0.0044)           | <b>-0.0055**</b><br>(0.0023)  | -0.0066<br>(0.0044)           |
| sCFO                       | <b>0.0081***</b><br>(0.0031)  | -0.0003<br>(0.0016)           | <b>0.0072***</b><br>(0.0026)  | -0.0005<br>(0.0016)           | <b>0.0135**</b><br>(0.0062)   | 0.0043<br>(0.0029)            | <b>0.0125**</b><br>(0.0055)   | 0.0042<br>(0.0029)            |
| SalesGrowth                | <b>-0.0189***</b><br>(0.0045) | <b>-0.0297***</b><br>(0.0048) | <b>-0.0176***</b><br>(0.0044) | <b>-0.0287***</b><br>(0.0048) | <b>-0.0158**</b><br>(0.0062)  | <b>-0.0156***</b><br>(0.0046) | <b>-0.0143**</b><br>(0.0062)  | <b>-0.0148***</b><br>(0.0047) |
| RDSales                    | <b>0.0089***</b><br>(0.0020)  | -0.0018<br>(0.0015)           | <b>0.0090***</b><br>(0.0020)  | -0.0016<br>(0.0015)           | <b>0.0152***</b><br>(0.0032)  | -0.0004<br>(0.0024)           | <b>0.0155***</b><br>(0.0032)  | -0.0003<br>(0.0024)           |
| CashFlowAt                 | <b>0.1415***</b><br>(0.0166)  | <b>0.1741***</b><br>(0.0180)  | <b>0.1368***</b><br>(0.0165)  | <b>0.1729***</b><br>(0.0180)  | <b>0.2109***</b><br>(0.0240)  | <b>0.1796***</b><br>(0.0257)  | <b>0.2012***</b><br>(0.0241)  | <b>0.1767***</b><br>(0.0259)  |
| DailyVola                  | <b>0.0001**</b><br>(0.0000)   | <b>-0.0001***</b><br>(0.0000) | <b>0.0001**</b><br>(0.0000)   | -0.0001<br>(0.0000)           | <b>0.0004***</b><br>(0.0001)  | <b>0.0001***</b><br>(0.0000)  | <b>0.0003***</b><br>(0.0001)  | 0.0001<br>(0.0001)            |
| Lagged_DV                  | <b>0.8605***</b><br>(0.0079)  | <b>0.6188***</b><br>(0.0133)  | <b>0.8571***</b><br>(0.0080)  | <b>0.6204***</b><br>(0.0134)  | <b>0.7310***</b><br>(0.0135)  | <b>0.2231***</b><br>(0.0193)  | <b>0.7250***</b><br>(0.0138)  | <b>0.2242***</b><br>(0.0193)  |
| Industry FE                | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               |
| Firm FE                    |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |
| Year FE                    | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           |                               |                               |
| Year-Quarter FE            |                               |                               | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           |
| N                          | 38,737                        | 38,737                        | 38,737                        | 38,737                        | 38,092                        | 38,092                        | 38,092                        | 38,092                        |
| R <sup>2</sup>             | 0.827                         | 0.432                         | 0.829                         | 0.432                         | 0.667                         | 0.100                         | 0.669                         | 0.100                         |
| Adj. R <sup>2</sup>        | 0.827                         | 0.412                         | 0.828                         | 0.411                         | 0.667                         | 0.069                         | 0.668                         | 0.067                         |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition: Clarity = -CEO FE; UncRes = call-level residual.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 5: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO  $\times$  Unrated Constraint

|                                    | (1)                           | (2)                           | (3)                           | (4)                           | (5)                           | (6)                           | (7)                           | (8)                           |
|------------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                                    | CashRatio                     |                               |                               |                               | CashRatio_Lead                |                               |                               |                               |
| CEO Clarity (DWZ Qtr-Exp, c)       | <b>-0.0114***</b><br>(0.0035) | <b>-0.0132***</b><br>(0.0052) | <b>-0.0104***</b><br>(0.0034) | <b>-0.0127***</b><br>(0.0053) | -0.0037<br>(0.0054)           | <b>-0.0186**</b><br>(0.0103)  | -0.0019<br>(0.0054)           | <b>-0.0187**</b><br>(0.0103)  |
| CEO Residual Unc. (DWZ Qtr-Exp, c) | <b>0.0021*</b><br>(0.0013)    | <b>0.0024**</b><br>(0.0013)   | <b>0.0021*</b><br>(0.0013)    | <b>0.0022**</b><br>(0.0013)   | <b>0.0044**</b><br>(0.0020)   | <b>0.0036**</b><br>(0.0017)   | <b>0.0044**</b><br>(0.0020)   | <b>0.0036**</b><br>(0.0017)   |
| CEO Pres Unc. (c)                  | 0.0000<br>(0.0013)            | -0.0005<br>(0.0013)           | 0.0000<br>(0.0013)            | -0.0004<br>(0.0013)           | 0.0003<br>(0.0025)            | 0.0009<br>(0.0022)            | 0.0000<br>(0.0025)            | 0.0009<br>(0.0022)            |
| Unrated                            | 0.0021<br>(0.0014)            | -0.0043<br>(0.0043)           | -0.0012<br>(0.0018)           | -0.0043<br>(0.0043)           | <b>0.0051**</b><br>(0.0024)   | -0.0073<br>(0.0066)           | -0.0019<br>(0.0032)           | -0.0074<br>(0.0066)           |
| UncRes (Qtr-Exp) $\times$ Unrated  | 0.0019<br>(0.0025)            | 0.0013<br>(0.0023)            | 0.0019<br>(0.0025)            | 0.0013<br>(0.0023)            | <b>0.0050*</b><br>(0.0038)    | 0.0023<br>(0.0033)            | <b>0.0050*</b><br>(0.0038)    | 0.0022<br>(0.0033)            |
| UncPre $\times$ Unrated            | 0.0014<br>(0.0026)            | -0.0014<br>(0.0029)           | 0.0017<br>(0.0026)            | -0.0014<br>(0.0029)           | -0.0001<br>(0.0046)           | -0.0015<br>(0.0046)           | 0.0004<br>(0.0046)            | -0.0014<br>(0.0046)           |
| Leverage                           | <b>-0.0084***</b><br>(0.0031) | -0.0170<br>(0.0104)           | <b>-0.0094***</b><br>(0.0034) | -0.0160<br>(0.0104)           | <b>-0.0210*</b><br>(0.0108)   | -0.0110<br>(0.0154)           | <b>-0.0240**</b><br>(0.0121)  | -0.0107<br>(0.0154)           |
| InAssets                           | 0.0003<br>(0.0003)            | <b>-0.0157***</b><br>(0.0031) | <b>-0.0016*</b><br>(0.0008)   | <b>-0.0157***</b><br>(0.0031) | <b>0.0016***</b><br>(0.0006)  | <b>-0.0300***</b><br>(0.0052) | <b>-0.0021*</b><br>(0.0013)   | <b>-0.0302***</b><br>(0.0053) |
| TobinsQ                            | <b>0.0039***</b><br>(0.0010)  | <b>0.0030**</b><br>(0.0015)   | <b>0.0034***</b><br>(0.0010)  | <b>0.0025*</b><br>(0.0014)    | <b>0.0049***</b><br>(0.0014)  | -0.0011<br>(0.0019)           | <b>0.0044***</b><br>(0.0015)  | -0.0011<br>(0.0020)           |
| ROA                                | <b>-0.0554***</b><br>(0.0139) | -0.0071<br>(0.0145)           | <b>-0.0535***</b><br>(0.0140) | -0.0065<br>(0.0146)           | <b>-0.1336***</b><br>(0.0211) | <b>-0.0761***</b><br>(0.0249) | <b>-0.1305***</b><br>(0.0213) | <b>-0.0773***</b><br>(0.0252) |
| Capex                              | <b>-0.1396***</b><br>(0.0175) | <b>-0.2564***</b><br>(0.0276) | <b>-0.1439***</b><br>(0.0176) | <b>-0.2536***</b><br>(0.0275) | <b>-0.2276***</b><br>(0.0263) | <b>-0.2870***</b><br>(0.0384) | <b>-0.2378***</b><br>(0.0262) | <b>-0.2865***</b><br>(0.0384) |
| DivDummy                           | <b>-0.0055***</b><br>(0.0016) | -0.0041<br>(0.0027)           | <b>-0.0056***</b><br>(0.0016) | -0.0041<br>(0.0026)           | <b>-0.0044*</b><br>(0.0025)   | -0.0076<br>(0.0053)           | <b>-0.0046*</b><br>(0.0026)   | -0.0077<br>(0.0053)           |
| sCFO                               | <b>0.0099***</b><br>(0.0030)  | 0.0017<br>(0.0016)            | <b>0.0094***</b><br>(0.0028)  | 0.0017<br>(0.0016)            | <b>0.0123*</b><br>(0.0069)    | 0.0041<br>(0.0032)            | <b>0.0116*</b><br>(0.0063)    | 0.0040<br>(0.0032)            |
| SalesGrowth                        | <b>-0.0128**</b><br>(0.0051)  | <b>-0.0267***</b><br>(0.0056) | <b>-0.0128**</b><br>(0.0051)  | <b>-0.0265***</b><br>(0.0056) | -0.0081<br>(0.0074)           | <b>-0.0127**</b><br>(0.0063)  | -0.0083<br>(0.0075)           | <b>-0.0127**</b><br>(0.0063)  |
| RDSales                            | <b>0.0086***</b><br>(0.0022)  | -0.0004<br>(0.0017)           | <b>0.0086***</b><br>(0.0022)  | -0.0003<br>(0.0017)           | <b>0.0154***</b><br>(0.0035)  | 0.0010<br>(0.0022)            | <b>0.0154***</b><br>(0.0035)  | 0.0011<br>(0.0022)            |
| CashFlowAt                         | <b>0.1220***</b><br>(0.0169)  | <b>0.1745***</b><br>(0.0188)  | <b>0.1204***</b><br>(0.0169)  | <b>0.1754***</b><br>(0.0189)  | <b>0.2021***</b><br>(0.0288)  | <b>0.1992***</b><br>(0.0317)  | <b>0.1968***</b><br>(0.0288)  | <b>0.1990***</b><br>(0.0317)  |
| DailyVola                          | 0.0000<br>(0.0000)            | <b>-0.0001**</b><br>(0.0000)  | -0.0000<br>(0.0000)           | <b>-0.0001**</b><br>(0.0000)  | <b>0.0003***</b><br>(0.0001)  | 0.0000<br>(0.0000)            | <b>0.0002***</b><br>(0.0001)  | 0.0000<br>(0.0001)            |
| Lagged_DV                          | <b>0.8737***</b><br>(0.0096)  | <b>0.6448***</b><br>(0.0166)  | <b>0.8713***</b><br>(0.0096)  | <b>0.6451***</b><br>(0.0166)  | <b>0.7471***</b><br>(0.0152)  | <b>0.2124***</b><br>(0.0230)  | <b>0.7408***</b><br>(0.0154)  | <b>0.2122***</b><br>(0.0229)  |
| Industry FE                        | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               |
| Firm FE                            |                               | Yes                           |                               | Yes                           |                               | Yes                           |                               | Yes                           |
| Year FE                            | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           |                               |                               |
| Year-Quarter FE                    |                               |                               | Yes                           | Yes                           |                               |                               | Yes                           | Yes                           |
| N                                  | 21,824                        | 21,824                        | 21,824                        | 21,824                        | 21,371                        | 21,371                        | 21,371                        | 21,371                        |
| $R^2$                              | 0.841                         | 0.453                         | 0.841                         | 0.453                         | 0.679                         | 0.094                         | 0.680                         | 0.094                         |
| Adj. $R^2$                         | 0.841                         | 0.423                         | 0.841                         | 0.423                         | 0.678                         | 0.046                         | 0.679                         | 0.044                         |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition under quarterly-expanding window (no-look-ahead).  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 6: H1.3 CEO 3-IV Decomp CFvol: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO  $\times$  HighCFvol (Han-Qiu 16q CV)

|                            | (1)                            | (2)                            | (3)                            | (4)                            | (5)                            | (6)                            | (7)                            | (8)                            |
|----------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
|                            |                                | CashRatio                      |                                |                                |                                | CashRatio <sub>lead</sub>      |                                |                                |
| CEO Clarity (DWZ, c)       | <b>-0.0088</b> ***<br>(0.0034) | -0.0059<br>(0.0047)            | <b>-0.0071</b> **<br>(0.0033)  | -0.0058<br>(0.0047)            | <b>-0.0096</b> **<br>(0.0052)  | <b>-0.0135</b> *<br>(0.0091)   | <b>-0.0078</b> *<br>(0.0052)   | <b>-0.0134</b> *<br>(0.0091)   |
| CEO Residual Unc. (DWZ, c) | <b>0.0021</b> **<br>(0.0011)   | <b>0.0017</b> *<br>(0.0010)    | <b>0.0022</b> **<br>(0.0011)   | <b>0.0017</b> **<br>(0.0010)   | <b>0.0025</b> **<br>(0.0015)   | 0.0016<br>(0.0013)             | <b>0.0024</b> *<br>(0.0015)    | 0.0015<br>(0.0014)             |
| CEO Pres Unc. (c)          | 0.0004<br>(0.0011)             | 0.0006<br>(0.0011)             | 0.0001<br>(0.0012)             | 0.0007<br>(0.0012)             | -0.0002<br>(0.0021)            | 0.0004<br>(0.0018)             | -0.0008<br>(0.0022)            | 0.0002<br>(0.0018)             |
| HighCFvol                  | -0.0005<br>(0.0012)            | -0.0022<br>(0.0016)            | <b>-0.0028</b> **<br>(0.0013)  | -0.0023<br>(0.0016)            | <b>0.0075</b> ***<br>(0.0019)  | 0.0021<br>(0.0025)             | <b>0.0049</b> **<br>(0.0020)   | 0.0021<br>(0.0025)             |
| UncRes $\times$ HighCFvol  | 0.0004<br>(0.0020)             | 0.0004<br>(0.0019)             | -0.0000<br>(0.0020)            | 0.0002<br>(0.0019)             | <b>0.0047</b> *<br>(0.0030)    | 0.0028<br>(0.0026)             | <b>0.0048</b> *<br>(0.0029)    | 0.0030<br>(0.0026)             |
| UncPre $\times$ HighCFvol  | <b>0.0048</b> **<br>(0.0022)   | 0.0002<br>(0.0024)             | <b>0.0049</b> **<br>(0.0022)   | 0.0002<br>(0.0024)             | <b>0.0099</b> ***<br>(0.0039)  | 0.0040<br>(0.0040)             | <b>0.0101</b> ***<br>(0.0039)  | 0.0041<br>(0.0040)             |
| Leverage                   | <b>-0.0127</b> ***<br>(0.0034) | <b>-0.0191</b> **<br>(0.0082)  | <b>-0.0130</b> **<br>(0.0036)  | <b>-0.0189</b> **<br>(0.0082)  | <b>-0.0273</b> ***<br>(0.0094) | -0.0116<br>(0.0121)            | <b>-0.0274</b> ***<br>(0.0097) | -0.0112<br>(0.0121)            |
| lnAssets                   | 0.0004<br>(0.0003)             | <b>-0.0174</b> ***<br>(0.0026) | <b>-0.0015</b> ***<br>(0.0006) | <b>-0.0169</b> ***<br>(0.0026) | <b>0.0014</b> ***<br>(0.0005)  | <b>-0.0236</b> ***<br>(0.0043) | -0.0013<br>(0.0010)            | <b>-0.0235</b> ***<br>(0.0044) |
| TobinsQ                    | <b>0.0044</b> **<br>(0.0008)   | <b>0.0037</b> ***<br>(0.0011)  | <b>0.0035</b> ***<br>(0.0008)  | <b>0.0034</b> ***<br>(0.0011)  | <b>0.0066</b> ***<br>(0.0013)  | 0.0016<br>(0.0017)             | <b>0.0056</b> ***<br>(0.0014)  | 0.0017<br>(0.0018)             |
| ROA                        | <b>-0.0641</b> ***<br>(0.0117) | -0.0148<br>(0.0123)            | <b>-0.0563</b> ***<br>(0.0116) | -0.0121<br>(0.0123)            | <b>-0.1405</b> ***<br>(0.0185) | <b>-0.0736</b> ***<br>(0.0205) | <b>-0.1330</b> ***<br>(0.0186) | <b>-0.0717</b> ***<br>(0.0206) |
| Capex                      | <b>-0.1692</b> ***<br>(0.0158) | <b>-0.3164</b> ***<br>(0.0251) | <b>-0.1779</b> ***<br>(0.0160) | <b>-0.3085</b> ***<br>(0.0249) | <b>-0.2443</b> ***<br>(0.0229) | <b>-0.3482</b> ***<br>(0.0363) | <b>-0.2565</b> ***<br>(0.0232) | <b>-0.3496</b> ***<br>(0.0364) |
| DivDummy                   | <b>-0.0057</b> ***<br>(0.0013) | <b>-0.0051</b> **<br>(0.0023)  | <b>-0.0054</b> ***<br>(0.0013) | <b>-0.0046</b> **<br>(0.0023)  | <b>-0.0048</b> ***<br>(0.0023) | <b>-0.0085</b> *<br>(0.0046)   | <b>-0.0049</b> **<br>(0.0023)  | <b>-0.0086</b> *<br>(0.0046)   |
| sCFO                       | <b>0.0087</b> **<br>(0.0043)   | 0.0004<br>(0.0017)             | <b>0.0078</b> **<br>(0.0037)   | 0.0003<br>(0.0017)             | <b>0.0128</b> *<br>(0.0071)    | 0.0041<br>(0.0031)             | <b>0.0121</b> *<br>(0.0066)    | 0.0040<br>(0.0030)             |
| SalesGrowth                | <b>-0.0203</b> ***<br>(0.0050) | <b>-0.0291</b> ***<br>(0.0057) | <b>-0.0192</b> ***<br>(0.0049) | <b>-0.0283</b> ***<br>(0.0057) | <b>-0.0187</b> ***<br>(0.0068) | <b>-0.0188</b> ***<br>(0.0055) | <b>-0.0172</b> **<br>(0.0067)  | <b>-0.0181</b> ***<br>(0.0056) |
| RDSales                    | <b>0.0063</b> ***<br>(0.0022)  | <b>-0.0031</b> **<br>(0.0015)  | <b>0.0063</b> ***<br>(0.0022)  | <b>-0.0029</b> **<br>(0.0015)  | <b>0.0153</b> ***<br>(0.0037)  | -0.0015<br>(0.0029)            | <b>0.0154</b> ***<br>(0.0036)  | -0.0014<br>(0.0029)            |
| CashFlowAt                 | <b>0.1476</b> ***<br>(0.0170)  | <b>0.1713</b> ***<br>(0.0179)  | <b>0.1382</b> ***<br>(0.0171)  | <b>0.1698</b> ***<br>(0.0179)  | <b>0.2228</b> ***<br>(0.0255)  | <b>0.1725</b> ***<br>(0.0262)  | <b>0.2093</b> ***<br>(0.0258)  | <b>0.1698</b> ***<br>(0.0264)  |
| DailyVola                  | <b>0.0001</b> ***<br>(0.0000)  | <b>-0.0001</b> ***<br>(0.0000) | <b>0.0001</b> ***<br>(0.0000)  | -0.0001<br>(0.0000)            | <b>0.0004</b> ***<br>(0.0001)  | <b>0.0001</b> ***<br>(0.0000)  | <b>0.0003</b> ***<br>(0.0001)  | 0.0001<br>(0.0001)             |
| Lagged_DV                  | <b>0.8664</b> ***<br>(0.0077)  | <b>0.6307</b> ***<br>(0.0133)  | <b>0.8619</b> ***<br>(0.0078)  | <b>0.6322</b> ***<br>(0.0133)  | <b>0.7262</b> ***<br>(0.0139)  | <b>0.2312</b> ***<br>(0.0205)  | <b>0.7200</b> ***<br>(0.0142)  | <b>0.2320</b> ***<br>(0.0205)  |
| Industry FE                | Yes                            |                                | Yes                            |                                | Yes                            |                                | Yes                            |                                |
| Firm FE                    |                                | Yes                            |                                | Yes                            |                                | Yes                            |                                | Yes                            |
| Year FE                    | Yes                            | Yes                            |                                |                                | Yes                            | Yes                            |                                |                                |
| Year-Quarter FE            |                                |                                | Yes                            | Yes                            |                                |                                | Yes                            | Yes                            |
| N                          | 40,795                         | 40,795                         | 40,795                         | 40,795                         | 38,671                         | 38,671                         | 38,671                         | 38,671                         |
| R <sup>2</sup>             | 0.825                          | 0.447                          | 0.826                          | 0.447                          | 0.657                          | 0.103                          | 0.659                          | 0.102                          |
| Adj. R <sup>2</sup>        | 0.824                          | 0.429                          | 0.826                          | 0.428                          | 0.657                          | 0.072                          | 0.658                          | 0.071                          |

Notes: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2018. Full panel coverage via Compustat Quarterly OCF Extended (1990-2002 backfill); Han-Qiu 16-trailing-quarter CFvol available from 1994-Q1 onwards in source data.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 7: H11: Political Risk and Language Uncertainty

|                             | (1)<br>Industry FE            |                                | (3)<br>Firm FE                |                               |
|-----------------------------|-------------------------------|--------------------------------|-------------------------------|-------------------------------|
|                             | UncResCEO                     | UncPreCEO                      | UncResCEO                     | UncPreCEO                     |
| Political Risk <sub>t</sub> | <b>0.0001</b> ***<br>(0.0000) | <b>0.0003</b> ***<br>(0.0000)  | <b>0.0001</b> ***<br>(0.0000) | <b>0.0002</b> ***<br>(0.0000) |
| UncQue                      | -0.0022<br>(0.0032)           | <b>0.0204</b> ***<br>(0.0045)  | -0.0014<br>(0.0037)           | <b>0.0081</b> ***<br>(0.0029) |
| NegCall                     | -0.0025<br>(0.0051)           | <b>0.1874</b> ***<br>(0.0141)  | -0.0036<br>(0.0073)           | <b>0.1477</b> ***<br>(0.0082) |
| lnAssets                    | -0.0002<br>(0.0004)           | <b>-0.0296</b> ***<br>(0.0039) | -0.0054<br>(0.0058)           | 0.0048<br>(0.0078)            |
| TobinsQ                     | -0.0005<br>(0.0009)           | -0.0007<br>(0.0036)            | -0.0008<br>(0.0022)           | 0.0001<br>(0.0023)            |
| ROA                         | <b>0.0291</b> **<br>(0.0145)  | 0.0477<br>(0.0318)             | <b>0.0540</b> **<br>(0.0264)  | 0.0346<br>(0.0230)            |
| CashRatio                   | 0.0019<br>(0.0077)            | -0.0354<br>(0.0357)            | 0.0206<br>(0.0240)            | 0.0265<br>(0.0287)            |
| DivDummy                    | -0.0031<br>(0.0021)           | -0.0139<br>(0.0118)            | -0.0108<br>(0.0074)           | <b>-0.0229</b> **<br>(0.0103) |
| FirmMat                     | 0.0012<br>(0.0013)            | <b>0.0021</b> **<br>(0.0010)   | 0.0027<br>(0.0041)            | 0.0006<br>(0.0007)            |
| EarnVol                     | 0.0078<br>(0.0077)            | -0.0072<br>(0.0157)            | 0.0056<br>(0.0117)            | -0.0104<br>(0.0132)           |
| UncPreCEO                   | -0.0051<br>(0.0034)           |                                | -0.0054<br>(0.0054)           |                               |
| Industry FE                 | Yes                           | Yes                            |                               |                               |
| Firm FE                     |                               |                                | Yes                           | Yes                           |
| Year FE                     | Yes                           | Yes                            | Yes                           | Yes                           |
| N                           | 44,497                        | 65,760                         | 44,497                        | 65,760                        |
| R <sup>2</sup>              | 0.003                         | 0.043                          | 0.003                         | 0.023                         |
| Adj. R <sup>2</sup>         | 0.002                         | 0.043                          | -0.030                        | -0.004                        |

Notes: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 8: H11-Lag2: Political Risk (2-Qtr Lag) and Language Uncertainty

|                             | (1)                         | (2)                            | (3)                          | (4)                           |
|-----------------------------|-----------------------------|--------------------------------|------------------------------|-------------------------------|
|                             | Industry FE                 |                                | Firm FE                      |                               |
|                             | UncResCEO                   | UncPreCEO                      | UncResCEO                    | UncPreCEO                     |
| PRisk <sub><i>t</i>-2</sub> | 0.0000<br>(0.0000)          | <b>0.0001</b> ***<br>(0.0000)  | 0.0000<br>(0.0000)           | <b>0.0000</b> ***<br>(0.0000) |
| UncQue                      | 0.0001<br>(0.0033)          | <b>0.0242</b> ***<br>(0.0047)  | 0.0010<br>(0.0038)           | <b>0.0099</b> ***<br>(0.0031) |
| NegCall                     | 0.0020<br>(0.0052)          | <b>0.2019</b> ***<br>(0.0145)  | 0.0019<br>(0.0074)           | <b>0.1609</b> ***<br>(0.0084) |
| lnAssets                    | -0.0001<br>(0.0005)         | <b>-0.0298</b> ***<br>(0.0040) | -0.0035<br>(0.0062)          | 0.0047<br>(0.0078)            |
| TobinsQ                     | -0.0005<br>(0.0010)         | -0.0002<br>(0.0037)            | -0.0011<br>(0.0023)          | -0.0002<br>(0.0023)           |
| ROA                         | <b>0.0293</b> *<br>(0.0152) | 0.0518<br>(0.0331)             | <b>0.0598</b> **<br>(0.0272) | <b>0.0464</b> *<br>(0.0247)   |
| CashRatio                   | 0.0064<br>(0.0076)          | -0.0400<br>(0.0368)            | 0.0170<br>(0.0242)           | 0.0233<br>(0.0292)            |
| DivDummy                    | -0.0031<br>(0.0022)         | -0.0142<br>(0.0121)            | <b>-0.0133</b> *<br>(0.0076) | <b>-0.0207</b> *<br>(0.0106)  |
| FirmMat                     | 0.0013<br>(0.0013)          | 0.0021<br>(0.0015)             | 0.0025<br>(0.0041)           | -0.0007<br>(0.0016)           |
| EarnVol                     | 0.0096<br>(0.0063)          | -0.0078<br>(0.0159)            | 0.0089<br>(0.0100)           | -0.0154<br>(0.0158)           |
| UncPreCEO                   | -0.0001<br>(0.0035)         |                                | -0.0003<br>(0.0056)          |                               |
| Industry FE                 | Yes                         | Yes                            |                              |                               |
| Firm FE                     |                             |                                | Yes                          | Yes                           |
| Year FE                     | Yes                         | Yes                            | Yes                          | Yes                           |
| N                           | 43,037                      | 63,022                         | 43,037                       | 63,022                        |
| R <sup>2</sup>              | 0.000                       | 0.035                          | 0.000                        | 0.015                         |
| Adj. R <sup>2</sup>         | -0.001                      | 0.034                          | -0.033                       | -0.013                        |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).



Table 9: H23: Product-Market Competition and Uncertainty Language

|                         | (1)                          | (2)                            | (3)                          | (4)                           |
|-------------------------|------------------------------|--------------------------------|------------------------------|-------------------------------|
|                         | Industry FE                  |                                | Firm FE                      |                               |
|                         | UncResCEO                    | UncPreCEO                      | UncResCEO                    | UncPreCEO                     |
| $z(\log(\text{TSIMM}))$ | 0.0008<br>(0.0014)           | <b>0.0304</b> ***<br>(0.0074)  | 0.0023<br>(0.0075)           | <b>0.0302</b> ***<br>(0.0093) |
| UncQue                  | 0.0011<br>(0.0062)           | <b>0.0398</b> ***<br>(0.0100)  | 0.0039<br>(0.0087)           | 0.0084<br>(0.0073)            |
| NegCall                 | 0.0020<br>(0.0068)           | <b>0.2340</b> ***<br>(0.0201)  | 0.0046<br>(0.0120)           | <b>0.1698</b> ***<br>(0.0140) |
| lnAssets                | -0.0007<br>(0.0007)          | <b>-0.0335</b> ***<br>(0.0038) | -0.0101<br>(0.0071)          | 0.0013<br>(0.0081)            |
| TobinsQ                 | 0.0001<br>(0.0013)           | -0.0007<br>(0.0037)            | 0.0009<br>(0.0025)           | -0.0017<br>(0.0025)           |
| ROA                     | 0.0233<br>(0.0197)           | <b>0.0681</b> **<br>(0.0314)   | 0.0343<br>(0.0308)           | 0.0241<br>(0.0254)            |
| CashRatio               | 0.0065<br>(0.0103)           | <b>-0.0803</b> **<br>(0.0360)  | 0.0055<br>(0.0287)           | 0.0022<br>(0.0306)            |
| DivDummy                | -0.0020<br>(0.0028)          | -0.0060<br>(0.0117)            | -0.0074<br>(0.0083)          | -0.0166<br>(0.0106)           |
| FirmMat                 | 0.0018<br>(0.0019)           | <b>0.0014</b> **<br>(0.0007)   | 0.0083<br>(0.0056)           | 0.0003<br>(0.0004)            |
| EarnVol                 | -0.0047<br>(0.0087)          | -0.0210<br>(0.0152)            | -0.0048<br>(0.0121)          | -0.0114<br>(0.0146)           |
| UncPreCEO               | <b>0.0111</b> **<br>(0.0044) |                                | <b>0.0230</b> **<br>(0.0093) |                               |
| Industry FE             | Yes                          | Yes                            |                              |                               |
| Firm FE                 |                              |                                | Yes                          | Yes                           |
| Year FE                 | Yes                          | Yes                            | Yes                          | Yes                           |
| N                       | 12,728                       | 18,492                         | 12,728                       | 18,492                        |
| $R^2$                   | 0.001                        | 0.050                          | 0.001                        | 0.021                         |
| Adj. $R^2$              | -0.002                       | 0.048                          | -0.098                       | -0.068                        |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Unit of observation: firm-fiscal-year.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 10: H24: US Economic Policy Uncertainty and Call Language Uncertainty

|                         | (1)<br>Industry + Cal. Year FE | (2)<br>UncPreCEO              | (3)<br>Firm + Cal. Year FE   | (4)<br>UncPreCEO             |
|-------------------------|--------------------------------|-------------------------------|------------------------------|------------------------------|
|                         | UncResCEO                      | UncPreCEO                     | UncResCEO                    | UncPreCEO                    |
| $\log(\text{US EPU})_t$ | <b>0.0124**</b><br>(0.0075)    | <b>0.0211***</b><br>(0.0074)  | <b>0.0123*</b><br>(0.0075)   | <b>0.0239***</b><br>(0.0076) |
| UncQue                  | 0.0011<br>(0.0035)             | <b>0.0164***</b><br>(0.0032)  | 0.0024<br>(0.0040)           | <b>0.0095***</b><br>(0.0028) |
| NegCall                 | 0.0006<br>(0.0056)             | <b>0.1247***</b><br>(0.0089)  | 0.0021<br>(0.0073)           | <b>0.1385***</b><br>(0.0079) |
| lnAssets                | 0.0009<br>(0.0007)             | <b>-0.0143***</b><br>(0.0020) | 0.0013<br>(0.0067)           | 0.0038<br>(0.0063)           |
| TobinsQ                 | -0.0001<br>(0.0011)            | 0.0007<br>(0.0019)            | -0.0000<br>(0.0027)          | 0.0007<br>(0.0018)           |
| ROA                     | <b>0.0281*</b><br>(0.0149)     | <b>0.0367**</b><br>(0.0168)   | <b>0.0677**</b><br>(0.0268)  | <b>0.0467**</b><br>(0.0191)  |
| CashRatio               | 0.0051<br>(0.0087)             | -0.0079<br>(0.0187)           | 0.0345<br>(0.0256)           | 0.0305<br>(0.0231)           |
| DivDummy                | <b>-0.0046***</b><br>(0.0009)  | -0.0074<br>(0.0061)           | <b>-0.0142**</b><br>(0.0072) | <b>-0.0157*</b><br>(0.0083)  |
| FirmMat                 | <b>0.0023*</b><br>(0.0012)     | 0.0009<br>(0.0008)            | 0.0005<br>(0.0047)           | -0.0011<br>(0.0012)          |
| EarnVol                 | <b>0.0268*</b><br>(0.0157)     | -0.0074<br>(0.0086)           | <b>0.0353**</b><br>(0.0176)  | -0.0083<br>(0.0119)          |
| UncPreCEO               | -0.0026<br>(0.0032)            |                               | -0.0011<br>(0.0045)          |                              |
| Lagged_DV               | <b>0.0202***</b><br>(0.0054)   | <b>0.5176***</b><br>(0.0131)  | <b>0.0187***</b><br>(0.0056) | <b>0.2626***</b><br>(0.0089) |
| Industry FE             | Yes                            | Yes                           |                              |                              |
| Firm FE                 |                                |                               | Yes                          | Yes                          |
| Year FE                 | Yes                            | Yes                           | Yes                          | Yes                          |
| N                       | 39,063                         | 60,503                        | 39,063                       | 60,503                       |
| $R^2$                   | 0.001                          | 0.292                         | 0.001                        | 0.084                        |
| Adj. $R^2$              | -0.000                         | 0.291                         | -0.035                       | 0.057                        |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 11: H24b: Global Economic Policy Uncertainty and Call Language Uncertainty

|                       | (1)                           | (2)                           | (3)                          | (4)                          |
|-----------------------|-------------------------------|-------------------------------|------------------------------|------------------------------|
|                       | Industry + Cal. Year FE       |                               | Firm + Cal. Year FE          |                              |
|                       | UncResCEO                     | UncPreCEO                     | UncResCEO                    | UncPreCEO                    |
| $\log(\text{GEPU})_t$ | <b>0.0181**</b><br>(0.0100)   | <b>0.0271***</b><br>(0.0107)  | <b>0.0187**</b><br>(0.0101)  | <b>0.0309***</b><br>(0.0108) |
| UncQue                | 0.0011<br>(0.0035)            | <b>0.0164***</b><br>(0.0032)  | 0.0024<br>(0.0039)           | <b>0.0095***</b><br>(0.0028) |
| NegCall               | 0.0006<br>(0.0056)            | <b>0.1247***</b><br>(0.0089)  | 0.0020<br>(0.0072)           | <b>0.1385***</b><br>(0.0079) |
| lnAssets              | 0.0009<br>(0.0007)            | <b>-0.0143***</b><br>(0.0020) | 0.0012<br>(0.0067)           | 0.0037<br>(0.0063)           |
| TobinsQ               | -0.0001<br>(0.0011)           | 0.0008<br>(0.0019)            | 0.0000<br>(0.0027)           | 0.0007<br>(0.0018)           |
| ROA                   | <b>0.0280*</b><br>(0.0149)    | <b>0.0367**</b><br>(0.0168)   | <b>0.0673**</b><br>(0.0267)  | <b>0.0462**</b><br>(0.0191)  |
| CashRatio             | 0.0051<br>(0.0087)            | -0.0080<br>(0.0188)           | 0.0345<br>(0.0256)           | 0.0304<br>(0.0231)           |
| DivDummy              | <b>-0.0047***</b><br>(0.0009) | -0.0074<br>(0.0061)           | <b>-0.0142**</b><br>(0.0072) | <b>-0.0156*</b><br>(0.0083)  |
| FirmMat               | <b>0.0023*</b><br>(0.0012)    | 0.0009<br>(0.0008)            | 0.0006<br>(0.0047)           | -0.0011<br>(0.0012)          |
| EarnVol               | <b>0.0269*</b><br>(0.0157)    | -0.0074<br>(0.0086)           | <b>0.0354**</b><br>(0.0176)  | -0.0082<br>(0.0119)          |
| UncPreCEO             | -0.0026<br>(0.0032)           |                               | -0.0011<br>(0.0045)          |                              |
| Lagged_DV             | <b>0.0202***</b><br>(0.0054)  | <b>0.5176***</b><br>(0.0131)  | <b>0.0187***</b><br>(0.0056) | <b>0.2626***</b><br>(0.0089) |
| Industry FE           | Yes                           | Yes                           |                              |                              |
| Firm FE               |                               |                               | Yes                          | Yes                          |
| Year FE               | Yes                           | Yes                           | Yes                          | Yes                          |
| N                     | 39,063                        | 60,503                        | 39,063                       | 60,503                       |
| $R^2$                 | 0.001                         | 0.292                         | 0.001                        | 0.084                        |
| Adj. $R^2$            | -0.000                        | 0.291                         | -0.035                       | 0.057                        |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

Table 12: H14c CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and 25-Day Post-Call Bid-Ask Spread Level ( $[0, +25]$ , the call day plus 25 trading days)

|                                | (1)                           | (2)                            | (3)                            | (4)                            | (5)                            | (6)                            | (7)                           | (8)                            | (9)                           | (10)                           | (11)                          | (12)                           |
|--------------------------------|-------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|-------------------------------|--------------------------------|-------------------------------|--------------------------------|-------------------------------|--------------------------------|
|                                | Spread <sub>25D,t</sub>       |                                |                                |                                |                                |                                | Spread <sub>25D,t+1</sub>     |                                |                               |                                |                               |                                |
| CEO Clarity (DWZ)              | -0.0477<br>(0.2145)           | 1.0617<br>(0.4277)             | 0.2122<br>(0.2227)             | 1.1056<br>(0.4260)             | 0.1644<br>(0.2103)             | 1.0547<br>(0.4174)             | -0.1886<br>(0.2370)           | 0.5998<br>(0.4004)             | -0.0922<br>(0.2399)           | 0.6828<br>(0.4055)             | -0.0431<br>(0.2206)           | 0.5722<br>(0.3729)             |
| CEO Residual Uncertainty (DWZ) | -0.0594<br>(0.1068)           | -0.0687<br>(0.1057)            | -0.0882<br>(0.1023)            | -0.1021<br>(0.1007)            | -0.0757<br>(0.0994)            | -0.0893<br>(0.0981)            | 0.0825<br>(0.1416)            | 0.0696<br>(0.1437)             | 0.0742<br>(0.1407)            | 0.0545<br>(0.1426)             | 0.0984<br>(0.1339)            | 0.0751<br>(0.1357)             |
| CEO Pres Uncertainty           | <b>0.1664**</b><br>(0.0946)   | <b>0.3496***</b><br>(0.1253)   | 0.0814<br>(0.0956)             | <b>0.2945***</b><br>(0.1229)   | 0.0108<br>(0.0919)             | <b>0.1644*</b><br>(0.1184)     | -0.0817<br>(0.0926)           | -0.0288<br>(0.1229)            | -0.1072<br>(0.0926)           | -0.0360<br>(0.1233)            | 0.0170<br>(0.0872)            | 0.1124<br>(0.1186)             |
| InAssets                       | <b>-0.8039***</b><br>(0.0490) | <b>-1.9098***</b><br>(0.2000)  | <b>-0.6339***</b><br>(0.0472)  | <b>-1.6825***</b><br>(0.2156)  | <b>-0.6352***</b><br>(0.0473)  | <b>-1.8277***</b><br>(0.2158)  | <b>-0.8218***</b><br>(0.0513) | <b>-1.5191***</b><br>(0.1951)  | <b>-0.7857***</b><br>(0.0505) | <b>-1.7292***</b><br>(0.2130)  | <b>-0.6568***</b><br>(0.0474) | <b>-1.4181***</b><br>(0.2017)  |
| TobinsQ                        | <b>-0.2861***</b><br>(0.0457) | <b>-0.4785***</b><br>(0.0900)  | <b>-0.2633***</b><br>(0.0470)  | <b>-0.5518***</b><br>(0.0963)  | <b>-0.2654***</b><br>(0.0451)  | <b>-0.5883***</b><br>(0.0956)  | <b>-0.2305***</b><br>(0.0467) | <b>-0.3215***</b><br>(0.0850)  | <b>-0.2285***</b><br>(0.0478) | <b>-0.4803***</b><br>(0.1002)  | <b>-0.2035***</b><br>(0.0433) | <b>-0.4289***</b><br>(0.0935)  |
| ROA                            | <b>-9.4469***</b><br>(1.0201) | <b>-11.6430***</b><br>(1.3867) | <b>-5.7059***</b><br>(0.9822)  | <b>-7.6844***</b><br>(1.2654)  | <b>-5.7511***</b><br>(0.9633)  | <b>-7.8517***</b><br>(1.2490)  | <b>-9.9256***</b><br>(1.0798) | <b>-12.8443***</b><br>(1.2992) | <b>-8.8744***</b><br>(1.0722) | <b>-11.9197***</b><br>(1.2732) | <b>-7.6375***</b><br>(0.9869) | <b>-10.5867***</b><br>(1.1921) |
| Leverage                       | <b>0.3875*</b><br>(0.2088)    | <b>1.8330***</b><br>(0.5143)   | -0.1126<br>(0.2434)            | 0.6850<br>(0.5013)             | -0.0542<br>(0.2264)            | 0.8026<br>(0.4954)             | 0.2924<br>(0.2224)            | 0.5282<br>(0.5516)             | 0.1338<br>(0.2248)            | 0.2221<br>(0.5596)             | 0.0592<br>(0.2111)            | 0.1928<br>(0.5258)             |
| Capex                          | -0.8001<br>(1.1109)           | 0.9540<br>(2.9126)             | <b>-2.3622**</b><br>(1.1044)   | -1.5271<br>(2.8061)            | <b>-2.0790*</b><br>(1.0845)    | -1.2897<br>(2.7998)            | <b>-1.6463*</b><br>(0.9620)   | -3.2121<br>(2.2020)            | <b>-2.1561**</b><br>(0.9823)  | <b>-4.3379*</b><br>(2.2446)    | -1.4770<br>(0.9011)           | -2.4356<br>(2.1565)            |
| DivDummy                       | 0.1043<br>(0.0893)            | 0.1429<br>(0.2020)             | <b>0.3329***</b><br>(0.0896)   | 0.2549<br>(0.1888)             | <b>0.3156***</b><br>(0.0855)   | 0.2723<br>(0.1867)             | 0.0240<br>(0.0887)            | 0.1007<br>(0.2095)             | 0.1012<br>(0.0889)            | 0.1454<br>(0.2074)             | 0.1145<br>(0.0825)            | 0.1739<br>(0.1975)             |
| sCFO                           | <b>0.6700**</b><br>(0.3405)   | <b>0.8405**</b><br>(0.3472)    | 0.4045<br>(0.4604)             | <b>0.6825*</b><br>(0.3789)     | 0.4084<br>(0.4126)             | <b>0.6882**</b><br>(0.3375)    | 0.0871<br>(0.1759)            | <b>0.3644*</b><br>(0.1877)     | 0.0040<br>(0.2174)            | 0.3165<br>(0.2055)             | -0.0840<br>(0.1937)           | 0.2479<br>(0.1738)             |
| Lagged_DV                      | <b>0.7547***</b><br>(0.0150)  | <b>0.6569***</b><br>(0.0186)   | <b>0.7140***</b><br>(0.0167)   | <b>0.6180***</b><br>(0.0199)   | <b>0.7310***</b><br>(0.0168)   | <b>0.6318***</b><br>(0.0208)   | <b>0.7540***</b><br>(0.0156)  | <b>0.6361***</b><br>(0.0185)   | <b>0.7360***</b><br>(0.0173)  | <b>0.6135***</b><br>(0.0205)   | <b>0.7676***</b><br>(0.0162)  | <b>0.6518***</b><br>(0.0197)   |
| StockPrice                     |                               |                                | -0.0011<br>(0.0014)            | -0.0033<br>(0.0024)            | 0.0002<br>(0.0013)             | 0.0019<br>(0.0024)             |                               |                                | -0.0001<br>(0.0013)           | <b>0.0101***</b><br>(0.0024)   | -0.0016<br>(0.0012)           | <b>0.0047**</b><br>(0.0022)    |
| Turnover                       |                               |                                | <b>-24.1012***</b><br>(1.7508) | <b>-23.8647***</b><br>(2.2119) | <b>-20.7840***</b><br>(1.7290) | <b>-20.3411***</b><br>(2.1791) |                               |                                | <b>-7.9396***</b><br>(1.5411) | <b>-5.5050***</b><br>(2.0904)  | <b>-7.1303***</b><br>(1.4699) | <b>-4.7073**</b><br>(1.9633)   |
| DailyVola                      |                               |                                | <b>0.1229***</b><br>(0.0072)   | <b>0.1393***</b><br>(0.0081)   | <b>0.1075***</b><br>(0.0082)   | <b>0.1257***</b><br>(0.0099)   |                               |                                | <b>0.0404***</b><br>(0.0062)  | <b>0.0520***</b><br>(0.0068)   | <b>0.0396***</b><br>(0.0069)  | <b>0.0441***</b><br>(0.0081)   |
| AbsSurpDec                     |                               |                                | <b>-0.0919***</b><br>(0.0265)  | -0.0241<br>(0.0300)            | <b>-0.0756***</b><br>(0.0253)  | -0.0093<br>(0.0286)            |                               |                                | -0.0050<br>(0.0278)           | 0.0119<br>(0.0304)             | -0.0174<br>(0.0261)           | -0.0012<br>(0.0285)            |
| Extended Controls              |                               |                                | Yes                            | Yes                            | Yes                            | Yes                            |                               |                                | Yes                           | Yes                            | Yes                           | Yes                            |
| Industry FE                    | Yes                           |                                | Yes                            |                                | Yes                            |                                | Yes                           |                                | Yes                           |                                | Yes                           |                                |
| Firm FE                        |                               | Yes                            |                                | Yes                            |                                | Yes                            |                               | Yes                            |                               | Yes                            |                               | Yes                            |
| Year FE                        | Yes                           | Yes                            | Yes                            | Yes                            |                                |                                | Yes                           | Yes                            | Yes                           | Yes                            |                               |                                |
| Year-Quarter FE                |                               |                                |                                |                                | Yes                            | Yes                            |                               |                                |                               |                                | Yes                           | Yes                            |
| N                              | 42,625                        | 42,625                         | 42,625                         | 42,625                         | 42,625                         | 42,625                         | 43,049                        | 43,049                         | 43,049                        | 43,049                         | 43,049                        | 43,049                         |
| R <sup>2</sup>                 | 0.703                         | 0.517                          | 0.721                          | 0.551                          | 0.730                          | 0.548                          | 0.700                         | 0.499                          | 0.702                         | 0.504                          | 0.723                         | 0.523                          |
| Adj. R <sup>2</sup>            | 0.703                         | 0.501                          | 0.721                          | 0.535                          | 0.730                          | 0.532                          | 0.700                         | 0.482                          | 0.702                         | 0.487                          | 0.723                         | 0.506                          |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ). Spread<sub>25D</sub> values rescaled by  $10^4$  (basis points) for readability; multiply coefficients by  $10^{-4}$  to recover raw fraction units.

Table 13: H18 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and SEC Conference-Call Comment Letter Receipt (LPM,  $[call_t, call_{t+1}]$  window)

|                                | (1)                           | (2)                           | (3)                           | (4)                           | (5)                           | (6)                           |
|--------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                                | CCCL                          |                               |                               |                               |                               |                               |
| CEO Clarity (DWZ)              | 0.0059<br>(0.0015)            | 0.0039<br>(0.0030)            | 0.0062<br>(0.0015)            | 0.0040<br>(0.0030)            | 0.0058<br>(0.0015)            | 0.0040<br>(0.0030)            |
| CEO Residual Uncertainty (DWZ) | -0.0003<br>(0.0009)           | -0.0003<br>(0.0009)           | -0.0003<br>(0.0009)           | -0.0003<br>(0.0009)           | -0.0004<br>(0.0009)           | -0.0004<br>(0.0009)           |
| CEO Pres Uncertainty           | 0.0008<br>(0.0008)            | <b>0.0014*</b><br>(0.0009)    | <b>0.0016**</b><br>(0.0007)   | <b>0.0014*</b><br>(0.0009)    | 0.0007<br>(0.0008)            | <b>0.0013*</b><br>(0.0009)    |
| InAssets                       | <b>-0.0006**</b><br>(0.0002)  | <b>0.0018*</b><br>(0.0011)    | 0.0001<br>(0.0001)            | 0.0017<br>(0.0010)            | <b>-0.0006**</b><br>(0.0003)  | <b>0.0019*</b><br>(0.0011)    |
| TobinsQ                        | -0.0003<br>(0.0003)           | 0.0002<br>(0.0004)            | -0.0001<br>(0.0003)           | 0.0002<br>(0.0004)            | -0.0002<br>(0.0003)           | 0.0004<br>(0.0004)            |
| ROA                            | 0.0006<br>(0.0034)            | 0.0012<br>(0.0052)            | -0.0014<br>(0.0052)           | -0.0006<br>(0.0062)           | -0.0006<br>(0.0051)           | -0.0004<br>(0.0062)           |
| Leverage                       | -0.0008<br>(0.0015)           | -0.0019<br>(0.0034)           | -0.0009<br>(0.0015)           | -0.0011<br>(0.0035)           | -0.0009<br>(0.0015)           | -0.0012<br>(0.0035)           |
| Capex                          | -0.0029<br>(0.0063)           | 0.0106<br>(0.0104)            | 0.0004<br>(0.0064)            | 0.0111<br>(0.0104)            | -0.0033<br>(0.0066)           | 0.0107<br>(0.0103)            |
| CashRatio                      | -0.0014<br>(0.0027)           | <b>-0.0075*</b><br>(0.0044)   | 0.0009<br>(0.0026)            | <b>-0.0080*</b><br>(0.0044)   | -0.0014<br>(0.0028)           | <b>-0.0082*</b><br>(0.0044)   |
| DivDummy                       | -0.0005<br>(0.0008)           | 0.0003<br>(0.0015)            | -0.0005<br>(0.0008)           | 0.0002<br>(0.0015)            | -0.0005<br>(0.0008)           | 0.0002<br>(0.0015)            |
| sCFO                           | -0.0007<br>(0.0005)           | -0.0020<br>(0.0024)           | -0.0004<br>(0.0004)           | -0.0019<br>(0.0024)           | -0.0007<br>(0.0005)           | -0.0019<br>(0.0024)           |
| Lagged_DV                      | <b>-0.0079***</b><br>(0.0008) | <b>-0.0409***</b><br>(0.0027) | <b>-0.0077***</b><br>(0.0007) | <b>-0.0410***</b><br>(0.0027) | <b>-0.0074***</b><br>(0.0008) | <b>-0.0405***</b><br>(0.0028) |
| SalesGrowth                    |                               |                               | -0.0002<br>(0.0008)           | -0.0013<br>(0.0011)           | -0.0001<br>(0.0008)           | -0.0011<br>(0.0011)           |
| RDSales                        |                               |                               | -0.0003<br>(0.0002)           | -0.0002<br>(0.0002)           | -0.0003<br>(0.0002)           | -0.0001<br>(0.0002)           |
| CashFlowAt                     |                               |                               | 0.0025<br>(0.0050)            | 0.0043<br>(0.0064)            | 0.0009<br>(0.0050)            | 0.0037<br>(0.0064)            |
| DailyVola                      |                               |                               | 0.0000<br>(0.0000)            | <b>-0.0000*</b><br>(0.0000)   | 0.0000<br>(0.0000)            | -0.0000<br>(0.0000)           |
| Extended Controls              |                               |                               | Yes                           | Yes                           | Yes                           | Yes                           |
| Industry FE                    | Yes                           |                               | Yes                           |                               | Yes                           |                               |
| Firm FE                        |                               | Yes                           |                               | Yes                           |                               | Yes                           |
| Year FE                        | Yes                           | Yes                           | Yes                           | Yes                           |                               |                               |
| Year-Quarter FE                |                               |                               |                               |                               | Yes                           | Yes                           |
| N                              | 44,113                        | 44,113                        | 44,096                        | 44,096                        | 44,096                        | 44,096                        |
| $R^2$                          | 0.001                         | 0.002                         | 0.000                         | 0.002                         | 0.001                         | 0.002                         |
| Adj. $R^2$                     | -0.000                        | -0.032                        | -0.001                        | -0.032                        | -0.001                        | -0.033                        |

Notes: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  (one-tailed for IVs,  $\beta > 0$ ; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual.  $R^2$  includes absorbed fixed effects (not within- $R^2$ ).

## **A Variable Definitions**

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

## A.1 Sample Manifest

| Name                                                                          | Description / Formula                                           | Source                       | Reference |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------|-----------|
| file_name, ceo_id,<br>ceo_name, gvkey,<br>ff12_code, ff12_name,<br>start_date | Master sample manifest with CEO, firm, and industry identifiers | outputs/1.4_AssembleManifest |           |

## A.2 Text/Linguistic Variables

| Name              | Description / Formula                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Source                                     | Reference                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------|
| ClarityCEO        | CEO clarity persistent-style component (DWZ Eqn 5).<br><i>Formula:</i> $\text{ClarityCEO}_i = -\widehat{\text{FE}}_i$ where $\widehat{\text{FE}}_i$ is the CEO- $i$ fixed-effect estimate from the panel regression $\text{UncAnsCEO}_{ic} = \alpha + \sum_i \text{FE}_i \cdot \mathbb{I}[\text{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ over the call-level Q&A uncertainty panel; controls $\mathbf{X}$ are the DWZ extended-controls vector. The sign flip on $\widehat{\text{FE}}_i$ makes higher ClarityCEO mean “CEO speaks more clearly” (lower mean uncertainty). Persistent CEO trait. | outputs/econometric/ceo_clarity_extended/  | <a href="#">Dzieliński et al. (2021)</a> (Section 4.4 Eqn 5, p.16)                     |
| ClarityCEO_QtrExp | ClarityCEO under within-tenure expanding window.<br><i>Formula:</i> As ClarityCEO, but the CEO fixed-effect regression is re-estimated on an expanding within-CEO window using only calls of CEO $i$ available up to and including focal call $c$ . Strictly look-ahead-free.                                                                                                                                                                                                                                                                                                                                | outputs/econometric/ceo_clarity_expanding/ | thesis QtrExp variant of <a href="#">Dzieliński et al. (2021)</a> Eqn 5                |
| NegCall           | Entire-call negative sentiment percentage.<br><i>Formula:</i> Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: ‘percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.’                                                                                                                                                                                                                                                                                                          | outputs/2_Textual_Analysis/2.2_Variables   | <a href="#">Dzieliński et al. (2021)</a> (Section 4.4, p.18; Appendix Table A.1, p.54) |
| UncAnsCEO         | CEO-only Q&A uncertainty percentage (input to ClarityCEO/UncResCEO residualization).<br><i>Formula:</i> $\text{CEO Q\&A uncertainty (\%)} = \text{UnctWordsAnsCEO} / \text{WordsAnsCEO}$ (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary ‘uncertainty’ list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field. Used in body only as the dependent variable inside the CEO fixed-effect residualization that produces ClarityCEO and UncResCEO.                                                                                                                     | outputs/2_Textual_Analysis/2.2_Variables   | <a href="#">Dzieliński et al. (2021)</a> (Section 4.2, Eqn 2, p.13)                    |
| UncPreCEO         | CEO-only presentation-segment uncertainty percentage (DWZ Eqn 1).<br><i>Formula:</i> $\text{CEO Presentation uncertainty (\%)} = \text{UnctWordsPreCEO} / \text{WordsPreCEO}$ . Uncertainty wordlist = LM 2011 (297 words).                                                                                                                                                                                                                                                                                                                                                                                  | outputs/2_Textual_Analysis/2.2_Variables   | <a href="#">Dzieliński et al. (2021)</a> (Section 4.2, Eqn 1, p.13)                    |
| UncQue            | Analyst Q&A uncertainty percentage (DWZ Eqn 3).<br><i>Formula:</i> $\text{Analyst Q\&A uncertainty (\%)} = \text{UnctWordsQue} / \text{WordsQue}$ . Uncertainty wordlist = LM 2011 (297 words). Used as $\Delta\text{UncQue}$ control in DWZ first-difference replication.                                                                                                                                                                                                                                                                                                                                   | outputs/2_Textual_Analysis/2.2_Variables   | <a href="#">Dzieliński et al. (2021)</a> (Section 4.2, Eqn 3, p.13)                    |

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| Name             | Description / Formula                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Source                                     | Reference                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------|
| UncResCEO        | CEO Q&A residual call-state uncertainty component (DWZ Eqn 4).<br><i>Formula:</i> $\hat{\varepsilon}_{ic}$ = the residual from the panel regression $\text{UncAnsCEO}_{ic} = \alpha + \sum_i \text{FE}_i \cdot \mathbb{I}[\text{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ that produces ClarityCEO. By OLS first-order conditions, the within-CEO mean of UncResCEO is exactly zero by construction. Captures call-level deviation in CEO Q&A uncertainty after stripping out the persistent CEO mean. | outputs/econometric/ceo_clarity_extended/  | <a href="#">Dzielninski et al. (2021)</a> (Section 4.4 Eqn 4, p.16)      |
| UncResCEO_QtrExp | UncResCEO under within-tenure expanding window.<br><i>Formula:</i> As UncResCEO, but the residualization regression is re-estimated on an expanding within-CEO window. Strictly look-ahead-free.                                                                                                                                                                                                                                                                                                                   | outputs/econometric/ceo_clarity_expanding/ | thesis QtrExp variant of <a href="#">Dzielninski et al. (2021)</a> Eqn 4 |

### A.3 Financial / Market Variables

| Name            | Description / Formula                                                                                                                                                                                                                                              | Source                                  | Reference                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------|
| BGTAvg_Amihud   | BGT 25-day symmetric average Amihud illiquidity (H7e).<br><i>Formula:</i> Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.                                             | .../variables/bgt_long_window_amihud.py | bgt2018 (window); F1D extension (symmetric average shape)            |
| BGTAvg_Spread   | BGT 25-day symmetric average bid-ask spread (H14e).<br><i>Formula:</i> Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018. | .../variables/bgt_long_window_spread.py | bgt2018 (window); F1D extension (symmetric average shape)            |
| BGTDelta_Amihud | BGT 25-day post-pre Amihud illiquidity change (H7d).<br><i>Formula:</i> Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.                                                      | .../variables/bgt_long_window_amihud.py | bgt2018 (window); F1D extension (delta shape)                        |
| BGTDelta_Spread | BGT 25-day post-pre bid-ask spread change (H14d).<br><i>Formula:</i> Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.                                                                                | .../variables/bgt_long_window_spread.py | bgt2018 (window); F1D extension (delta shape)                        |
| BGTLevel_Amihud | Compute BGT (2018) long-window Amihud illiquidity around earnings calls.<br><i>Formula:</i> CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/bgt_long_window_amihud.py)                                                                           | CRSPviaget_raw_daily_data               | Bushee, Gow & Taylor (2018, JAR) 56(1):85-121                        |
| BGTLevel_Spread | Compute BGT (2018) long-window closing bid-ask spread around earnings calls.<br><i>Formula:</i> CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)                                                     | CRSPviaget_raw_daily_data               | Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula] |
| Capex           | Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)                                                                                        | Compustat/capxy_Q4/atq                  | Bates, Kahle & Stulz (2009, JF)                                      |
| CashFlowAt      | Build CashFlow = oancfy / atq from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)                                                                                                       | Compustat/oancfy/atq                    | Bates, Kahle & Stulz (2009, JF)                                      |

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| Name                 | Description / Formula                                                                                                                                                                                                                                                                                                                              | Source                             | Reference                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CashRatio            | Build CashRatio = cheq / atq from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)                                                                                                                                                                                      | Compustat/cheq/atq                 | <a href="#">Bates et al. (2009)</a> (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)                                      |
| ChangDebtChoice      | ChangDebtChoice (description not in module docstring)<br><i>Formula:</i> see module: src/f1d/shared/variables/_compustat_engine.py                                                                                                                                                                                                                 | .../variables/_compustat_engine.py | cdh2006                                                                                                                                                              |
| ChangExternalFunding | ChangExternalFunding (description not in module docstring)<br><i>Formula:</i> see module: src/f1d/shared/variables/_compustat_engine.py                                                                                                                                                                                                            | .../variables/_compustat_engine.py | cdh2006                                                                                                                                                              |
| DailyVola            | Build Volatility from raw CRSP daily stock files via CRSPEngine.<br><i>Formula:</i> CRSP/RET (see module: src/f1d/shared/variables/volatility.py)                                                                                                                                                                                                  | CRSP/RET                           | dwz2021                                                                                                                                                              |
| DebtToCapital        | Build DebtToCapital = total debt / (equity + total debt).<br><i>Formula:</i> Compustat/dlcq,dlttq,seqq (see module: src/f1d/shared/variables/debt_to_capital.py)                                                                                                                                                                                   | Compustat/dlcq,dlttq,seqq          | rz1995                                                                                                                                                               |
| DivDummy             | Build DivDummy = (dvy $\neq$ 0).astype(float) from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/dvy $\neq$ 0 (see module: src/f1d/shared/variables/dividend_payer.py)                                                                                                                                                                | Compustat/dvy $\neq$ 0             | <a href="#">Bates et al. (2009)</a> (Section IV, p.2000; dividend payout dummy based on common dividends)                                                            |
| DivPayerQ            | Build DivPayerQ = (dvpsxq $\neq$ 0).astype(float) from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/dvpsxq $\neq$ 0 (quarterly ex-date) (see module: src/f1d/shared/variables/dividend_payer_quarterly.py)                                                                                                                           | Compustat/dvpsxq $\neq$ 0          | Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog                                                                                                     |
| EarnVol              | Builder for Earnings Volatility (earnings_volatility) variable.<br><i>Formula:</i> Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)                                                                                                                                                                                         | Compustat                          | <a href="#">Duong, Do, and Do (2025)</a> (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)                                                         |
| EquityDelayCon       | Hoberg-Maksimovic (2015) equity-market financing constraint index (H22).<br><i>Formula:</i> Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase 'equitydelaycon'; CamelCase name is the spec convention. See docs/provenance/h22.md for merge details. | inputs/Hoberg_Maksimovic/          | hm2015                                                                                                                                                               |
| FirmMat              | Builder for Firm Maturity (firm_maturity) variable.<br><i>Formula:</i> Compustat (see module: src/f1d/shared/variables/firm_maturity.py)                                                                                                                                                                                                           | Compustat                          | <a href="#">Biddle, Hilary, and Verdi (2009)</a> (Appendix A, p.130; years since first CRSP appearance); alt <a href="#">Leary and Roberts (2010)</a> (App.~A p.354) |
| Leverage             | Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only).<br><i>Formula:</i> Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)                                                                                                                                                                                 | Compustat/dlcq,dlttq,atq           | bks2009, rz1995                                                                                                                                                      |
| lnAssets             | Build Size = ln(atq) from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/atq (see module: src/f1d/shared/variables/size.py)                                                                                                                                                                                                            | Compustat/atq                      | dwz2021                                                                                                                                                              |
| Loss                 | Builder for Loss (H5 Control Variable).<br><i>Formula:</i> Compustat ibq $\neq$ 0 (see module: src/f1d/shared/variables/loss_dummy.py)                                                                                                                                                                                                             | Compustatibq $\neq$ 0              | <a href="#">Biddle et al. (2009)</a> (Appendix A, p.131; indicator for negative net income before extraordinary items)                                               |

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| Name                | Description / Formula                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Source                                   | Reference                                                                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| PayoutRatio_q       | Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat.<br><i>Formula:</i> Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/f1d/shared/variables/payout_ratio_quarterly.py)                                                                                                                                                                                                                                                                                                                                                                                                                                  | Compustat/                               | dds2004                                                                                                                                           |
| PRisk               | Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter).<br><i>Formula:</i> Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)                                                                                                                                                                                                                                                                                                                                                                                                                                      | Hassanetal.                              | hhl2019                                                                                                                                           |
| RDSales             | Build RDSales = xrdq / atq from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/xrdq.atq (see module: src/f1d/shared/variables/rd_intensity.py)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Compustat/xrdq.atq                       | bks2009, jjl2021                                                                                                                                  |
| RepurchaseIntensity | Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat.<br><i>Formula:</i> Compustat/quarterly_prstkcy/lagged_atq (see module: src/f1d/shared/variables/repurchase_intensity.py)                                                                                                                                                                                                                                                                                                                                                                                                                        | Compustat/quarterly_prstkcy/lagged_atq   | Thesis Advisor Suggestion                                                                                                                         |
| ROA                 | Build ROA = iby_annual / avg_assets from Compustat quarterly data.<br><i>Formula:</i> Compustat/iby.atq (see module: src/f1d/shared/variables/roa.py)                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Compustat/iby.atq                        | <a href="#">Wang (2020)</a> , dwz2021                                                                                                             |
| SalesGrowth         | Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel).<br><i>Formula:</i> Compustat/saleq_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Compustat/saleq_Q4_YoY                   | <a href="#">Biddle et al. (2009)</a> (Section 3.2, p.117; pct change in sales)                                                                    |
| sCFO                | Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey.<br><i>Formula:</i> Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)                                                                                                                                                                                                                                                                                                                                                                                                                            | Compustat/oancfy/atq_{t-1}rolli<br>ngstd | <a href="#">Biddle et al. (2009)</a> (Appendix A, p.130; 5-year rolling std of CFO/avg assets)                                                    |
| StockPrice          | Compute stock price at earnings call date (VECTORIZED).<br><i>Formula:</i> CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CRSPviaget_raw_daily_data                | <a href="#">Amiram, Owens, and Rozenbaum (2016)</a> (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)                   |
| SurpDec             | Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter.<br><i>Formula:</i> Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009). | .../variables/earnings_surprise.py       | <a href="#">Dzielinski et al. (2021)</a> (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)          |
| TobinsQ             | Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data.<br><i>Formula:</i> Compustat/(atq+cshoq*prccq-ceq)/atq (see module: src/f1d/shared/variables/tobins_q.py)                                                                                                                                                                                                                                                                                                                                                                                                                                 | Compustat/                               | Bates, Kahle & Stulz (2009, JF)                                                                                                                   |
| Turnover            | Compute share turnover at earnings call date (VECTORIZED).<br><i>Formula:</i> CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CRSPviaget_raw_daily_data                | <a href="#">Amihud (2002)</a> (Section 2.1, p.33; alt rolling-window def in <a href="#">Chang, Dasgupta, and Hilary (2006)</a> Appendix B p.3045) |

## A.4 Econometric / Indicator Variables

| Name                  | Description / Formula                                                                                                                                                                                                                                                                 | Source  | Reference                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------|
| BelowIG               | Below-IG credit rating subsample indicator (H1.2).<br><i>Formula:</i> Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).                           | runtime | thesis (H1.2 rating subsample)                                      |
| HighCFvol             | High cash-flow-volatility moderator (H1.3).<br><i>Formula:</i> Binary = 1 if firm-quarter Han-Qiu (2007) cash-flow-volatility (16-quarter rolling std of OCF /  mean  of OCF) is above the within-industry-year median; 0 otherwise.                                                  | runtime | <a href="#">Han and Qiu (2007)</a> ; thesis (H1.3 split convention) |
| HighTSIMM             | High product-market-similarity indicator (H1.1).<br><i>Formula:</i> Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.                                                                                         | runtime | hp2016 (TSIMM construction); thesis (H1.1 split convention)         |
| Lagged_DV             | Generic lagged-DV control (1Q lag of suite-specific DV).<br><i>Formula:</i> 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.                                                    | runtime | thesis convention (per feedback_lagged.dv.md)                       |
| Unrated               | No-rating subsample indicator (H1.2).<br><i>Formula:</i> Binary = 1 if firm has no S&P rating at call date; 0 otherwise.                                                                                                                                                              | runtime | thesis (H1.2 rating subsample)                                      |
| z_log_TotalSimilarity | z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1).<br><i>Formula:</i> z-score of $\log(1 + \text{TotalSimilarity})$ computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities $\times 100$ ). | runtime | hp2016                                                              |

## A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

| Name                  | Description / Formula                                                                                                                                      | Source  | Reference                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------|
| AbsSurpDec            | Absolute earnings surprise quintile magnitude.<br><i>Formula:</i> abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5. | runtime | thesis (magnitude control independent of sign)            |
| BGTAvg_Amihud_lead1   | 1Q lead of BGTAvg_Amihud.<br><i>Formula:</i> 1-quarter lead of BGTAvg_Amihud                                                                               | runtime | bgt2018 (window); F1D extension (symmetric average shape) |
| BGTAvg_Spread_lead1   | 1Q lead of BGTAvg_Spread.<br><i>Formula:</i> 1-quarter lead of BGTAvg_Spread                                                                               | runtime | bgt2018 (window); F1D extension (symmetric average shape) |
| BGTDelta_Amihud_lead1 | 1Q lead of BGTDelta_Amihud.<br><i>Formula:</i> 1-quarter lead of BGTDelta_Amihud                                                                           | runtime | bgt2018 (window); F1D extension (delta shape)             |
| BGTDelta_Spread_lead1 | 1Q lead of BGTDelta_Spread.<br><i>Formula:</i> 1-quarter lead of BGTDelta_Spread                                                                           | runtime | bgt2018 (window); F1D extension (delta shape)             |
| BGTLevel_Amihud_lead1 | 1Q lead of BGTLevel_Amihud.<br><i>Formula:</i> 1-quarter lead of BGTLevel_Amihud                                                                           | runtime | bgt2018                                                   |

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| Name                      | Description / Formula                                                                      | Source  | Reference                                                                                                         |
|---------------------------|--------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------|
| BGTLevel_Spread_lead1     | 1Q lead of BGTLevel_Spread.<br><i>Formula:</i> 1-quarter lead of BGTLevel_Spread           | runtime | bgt2018, lee2016                                                                                                  |
| Capex_lead                | 1Q lead of Capex.<br><i>Formula:</i> 1-quarter lead of Capex                               | runtime | Bates, Kahle & Stulz (2009, JF)                                                                                   |
| Capex_lead2               | 2Q lead of Capex.<br><i>Formula:</i> 2-quarter lead of Capex                               | runtime | Bates, Kahle & Stulz (2009, JF)                                                                                   |
| Capex_lead3               | 3Q lead of Capex.<br><i>Formula:</i> 3-quarter lead of Capex                               | runtime | Bates, Kahle & Stulz (2009, JF)                                                                                   |
| Capex_lead4               | 4Q lead of Capex.<br><i>Formula:</i> 4-quarter lead of Capex                               | runtime | Bates, Kahle & Stulz (2009, JF)                                                                                   |
| CashRatio_lead            | 1Q lead of CashRatio.<br><i>Formula:</i> 1-quarter lead of CashRatio                       | runtime | see base var                                                                                                      |
| ChangDebtChoice_lead      | 1Q lead of ChangDebtChoice.<br><i>Formula:</i> 1-quarter lead of ChangDebtChoice           | runtime | cdh2006                                                                                                           |
| ChangExternalFunding_lead | 1Q lead of ChangExternalFunding.<br><i>Formula:</i> 1-quarter lead of ChangExternalFunding | runtime | cdh2006                                                                                                           |
| DebtToCapital_lead        | 1Q lead of DebtToCapital.<br><i>Formula:</i> 1-quarter lead of DebtToCapital               | runtime | rz1995                                                                                                            |
| DeltaILLIQ_lead1          | 1Q lead of DeltaILLIQ.<br><i>Formula:</i> 1-quarter lead of DeltaILLIQ                     | runtime | amihud02                                                                                                          |
| DISP_lag                  | 1Q lag of DISP.<br><i>Formula:</i> 1-quarter lag of DISP                                   | runtime | <a href="#">Wang (2020)</a>                                                                                       |
| DISP_lead                 | 1Q lead of DISP.<br><i>Formula:</i> 1-quarter lead of DISP                                 | runtime | <a href="#">Wang (2020)</a>                                                                                       |
| DivPayerQ_lead1           | 1Q lead of DivPayerQ.<br><i>Formula:</i> 1-quarter lead of DivPayerQ                       | runtime | Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent) |
| DSPREAD_lead1             | 1Q lead of DSPREAD.<br><i>Formula:</i> 1-quarter lead of DSPREAD                           | runtime | lee2016                                                                                                           |
| EquityDelayCon_lead       | 1Q lead of EquityDelayCon.<br><i>Formula:</i> 1-quarter lead of EquityDelayCon             | runtime | hm2015                                                                                                            |
| Leverage_lead             | 1Q lead of Leverage.<br><i>Formula:</i> 1-quarter lead of Leverage                         | runtime | bks2009, rz1995                                                                                                   |

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| Name                         | Description / Formula                                                                                                                                                                                                                                         | Source  | Reference                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------|
| PayoutRatio_q_lead_qtr       | 1Q lead of PayoutRatio_q (qtr-aligned).<br><i>Formula:</i> 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)                                                                                                                                         | runtime | dds2004                                                                                                 |
| PostCallAmihud               | Panel-time post-call Amihud illiquidity level (H7b).<br><i>Formula:</i> Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.    | runtime | thesis (H7b panel-time post-call construction)                                                          |
| PostCallAmihud_lead1         | 1Q lead of PostCallAmihud.<br><i>Formula:</i> 1-quarter lead of PostCallAmihud                                                                                                                                                                                | runtime | thesis (H7b panel-time post-call construction)                                                          |
| PostCallSpread               | Panel-time post-call bid-ask spread level (H14b).<br><i>Formula:</i> Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering. | runtime | thesis (H14b panel-time post-call construction)                                                         |
| PostCallSpread_lead1         | 1Q lead of PostCallSpread.<br><i>Formula:</i> 1-quarter lead of PostCallSpread                                                                                                                                                                                | runtime | thesis (H14b panel-time post-call construction)                                                         |
| PRisk_lag                    | 1Q lag of PRisk.<br><i>Formula:</i> 1-quarter lag of PRisk                                                                                                                                                                                                    | runtime | hhlt2019                                                                                                |
| PRisk_lag2                   | 2Q lag of PRisk.<br><i>Formula:</i> 2-quarter lag of PRisk                                                                                                                                                                                                    | runtime | hhlt2019                                                                                                |
| RDSales_lead                 | 1Q lead of RDSales.<br><i>Formula:</i> 1-quarter lead of RDSales                                                                                                                                                                                              | runtime | bks2009, jjl2021                                                                                        |
| RepurchaseIntensity_lead_qtr | 1Q lead of RepurchaseIntensity (qtr-aligned).<br><i>Formula:</i> 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)                                                                                                                             | runtime | Thesis Advisor Suggestion                                                                               |
| ClarityCEO_c                 | Mean-centered ClarityCEO.<br><i>Formula:</i> ClarityCEO minus its sample mean (mean-centered for interaction interpretability)                                                                                                                                | runtime | thesis (interaction interpretability)                                                                   |
| ClarityCEO_QtrExp_c          | Mean-centered ClarityCEO_QtrExp.<br><i>Formula:</i> ClarityCEO_QtrExp minus its sample mean                                                                                                                                                                   | runtime | thesis (interaction interpretability)                                                                   |
| Post                         | DiD post-event-period indicator (Phase E sudden-death).<br><i>Formula:</i> Binary = 1 if call-quarter is on or after the death-event quarter for the treated firm (and on or after the matched event-quarter for the matched-control firm); 0 otherwise.      | runtime | <a href="#">Ghafoor, Yousaf, and Li (2023)</a>                                                          |
| Treated                      | DiD treated-firm indicator (Phase E sudden-death).<br><i>Formula:</i> Binary = 1 if firm experienced a CEO sudden-death event in 2002–2018; 0 if PSM-matched control firm.                                                                                    | runtime | <a href="#">Bennedsen, Pérez-González, and Wolfenzon (2020)</a> ; <a href="#">Ghafoor et al. (2023)</a> |
| Treated_x_Post               | DiD ATT term (Phase E sudden-death).<br><i>Formula:</i> Treated × Post. The coefficient on this product is the average treatment effect on the treated.                                                                                                       | runtime | <a href="#">Bertrand, Duflo, and Mullainathan (2004)</a> ; <a href="#">Ghafoor et al. (2023)</a>        |

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| Name                         | Description / Formula                                                                                                        | Source  | Reference                             |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------|
| UncPreCEO_c                  | Mean-centered UncPreCEO.<br><i>Formula:</i> UncPreCEO minus its sample mean (mean-centered for interaction interpretability) | runtime | thesis (interaction interpretability) |
| UncPreCEO_c.x.HighCFvol      | Interaction: UncPreCEO_c $\times$ HighCFvol (H1.3).<br><i>Formula:</i> Product: UncPreCEO_c * HighCFvol                      | runtime | thesis (CFvol moderator interaction)  |
| UncPreCEO_c.x.Unrated        | Interaction: UncPreCEO_c $\times$ Unrated (HFC).<br><i>Formula:</i> Product: UncPreCEO_c * Unrated                           | runtime | thesis (HFC interaction)              |
| UncResCEO_c                  | Mean-centered UncResCEO.<br><i>Formula:</i> UncResCEO minus its sample mean (mean-centered for interaction interpretability) | runtime | thesis (interaction interpretability) |
| UncResCEO_c.x.HighCFvol      | Interaction: UncResCEO_c $\times$ HighCFvol (H1.3).<br><i>Formula:</i> Product: UncResCEO_c * HighCFvol                      | runtime | thesis (CFvol moderator interaction)  |
| UncResCEO_c.x.Unrated        | Interaction: UncResCEO_c $\times$ Unrated (HFC).<br><i>Formula:</i> Product: UncResCEO_c * Unrated                           | runtime | thesis (HFC interaction)              |
| UncResCEO_QtrExp_c           | Mean-centered UncResCEO_QtrExp.<br><i>Formula:</i> UncResCEO_QtrExp minus its sample mean                                    | runtime | thesis (interaction interpretability) |
| UncResCEO_QtrExp_c.x.Unrated | Interaction: UncResCEO_QtrExp_c $\times$ Unrated (HFC QtrExp).<br><i>Formula:</i> Product: UncResCEO_QtrExp_c * Unrated      | runtime | thesis (HFC QtrExp interaction)       |

## A.6 Stage 9

| Name       | Description / Formula                                                                                                                                                                                                                                     | Source                                       | Reference                    |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------|
| CCCL       | SEC comment letter received (binary, forward-looking)<br><i>Formula:</i> 1 if SEC comment letter received in next quarter, 0 otherwise                                                                                                                    | inputs/EDGAR_CommentLetters                  | cdm2013                      |
| DeltaILLIQ | Change in Amihud illiquidity around earnings call<br><i>Formula:</i> PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days                                                                                                                            | inputs/CRSP_DSF                              | amihud02                     |
| DISP       | Analyst forecast dispersion (Wang 2020)<br><i>Formula:</i> SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior                                                                                                                             | inputs/tr_ibes                               | <a href="#">Wang (2020)</a>  |
| DSPREAD    | Change in closing bid-ask spread around earnings call<br><i>Formula:</i> mean(RelSpread[+1,+3]) - mean(RelSpread[-3,-1]) where RelSpread = 2*(ASK-BID)/(ASK+BID)                                                                                          | inputs/CRSP_DSF                              | lee2016                      |
| GEPU_log   | Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-\$ GDP weights)<br><i>Formula:</i> log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start.date | inputs/EconomicPolicyUncertaintyIndex/Global | <a href="#">Davis (2016)</a> |

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| Name            | Description / Formula                                                                                                                                                                                                                | Source                                   | Reference                           |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------|
| GPR_log         | Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version)<br><i>Formula:</i> log(GPR) — recent headline Geopolitical Risk index, matched by calendar month of call start_date | inputs/matteoiacoviello/GPR_Global       | ci2022                              |
| SEC_Letters_fwd | SEC comment letter count (forward-looking)<br><i>Formula:</i> Count of SEC-originated letters (UPLOAD type) in next calendar quarter                                                                                                 | inputs/EDGAR_CommentLetters              | cdm2013 (thesis extension to count) |
| TotalSimilarity | Hoberg-Phillips total product similarity<br><i>Formula:</i> Sum of pairwise cosine similarities from 10-K product descriptions                                                                                                       | inputs/TNIC                              | hp2016                              |
| US_EPU_log      | Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed)<br><i>Formula:</i> log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date                               | inputs/EconomicPolicyUncertaintyIndex/US | bbd2016                             |